GEOMETRIC SINGULARITIES OF REGULAR SURFACES WITH NEF ANTI-CANONICAL DIVISORS OVER IMPERFECT FIELDS

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ABSTRACT. Let S be a regular projective surface over a field k of characteristic $p > 0$, with $H^0(S, \mathcal{O}_S) = k$ and $-K_S$ nef. We prove that S is geometrically integral over k when $p \geq 7$, and we also find an example of S that is not geometrically integral when $p = 5$.

1. INTRODUCTION

Let X be a regular projective variety over a field k of characteristic $p > 0$. When k is imperfect, X is not necessarily smooth. Such varieties arise naturally as generic fibers of fibrations $f\colon \mathcal{X} \to T$ between smooth varieties with $k = K(T)$. The singularities that X acquires after base change to $\overline{k}$—its *geometric singularities*—can be quite bad. For instance, over $k = \mathbb{F}_p(s_1, \cdots, s_n)$, the hypersurface defined by the equation $X_0^p + s_1 X_1^p + \cdots + s_n X_n^p = 0$ in $\mathbb{P}^n_k$ is regular, while its base change to $\overline{k}$ is a p-fold hyperplane. In this paper, we focus on varieties with $-K_X$ being nef, such as the generic fibers of Mori fiber spaces or of Iitaka fibrations of K_X. Such varieties play an important role in the classification of varieties in characteristic p, and understanding their geometric singularities is a fundamental problem. It is expected that they have mild geometric singularities when p is large; for example, the anticanonical divisor of the above hypersurface is nef if and only if $p \leq n + 1$. The following cases have been extensively studied:

- $\dim X = 1$ ([Que71, Sch10, Tan21, Sch22]): if $-K_X$ is ample, then X is a conic in $\mathbb{P}^2$, which is smooth when $p \geq 3$; and if $K_X \equiv 0$, then X is smooth when $p \geq 5$, while for $p = 2, 3$ such curves are completely classified [Sch22].
- $\dim X = 2$ with $-K_X$ ample ([MS03, PW22, FS20, BT22, BM24, BT24] among others): X is geometrically normal when $p \geq 5$ ([PW22, Theorem 1.5] or [BT22, Theorem 3.7]), and geometrically regular when $p \geq 11$ ([BT22, Proposition 5.2]).

Question 1.1. *Let S be a regular projective surface over k with $-K_S$ nef and $H^0(S, \mathcal{O}_S) = k$.*

(Q1) *Is there a number N_1 such that every such S in characteristic $p > N_1$ is geometrically integral?*

(Q2) *Is there a number N_2 such that every such S in characteristic $p > N_2$ is geometrically normal?*

When $p = 2$ or 3, it is known that S can fail to be geometrically integral. Let k_0 be a perfect field of characteristic p and let $k = k_0(a_0, \ldots, a_9)$ with the a_i algebraically independent. The following surfaces are regular and geometrically non-reduced:

$$\begin{aligned} -K_S \text{ ample}: &\quad \textstyle\sum_{i=0}^3 a_i X_i^p = 0 \ \subset\ \mathbb{P}^3_k, \\ K_S \sim 0,\ p = 2: &\quad \textstyle\sum_{i=0}^3 a_i X_i^4 = 0 \ \subset\ \mathbb{P}^3_k, \\ K_S \sim 0,\ p = 3: &\quad \textstyle\sum_{i=0}^4 a_i X_i^3 = \sum_{i=0}^4 a_{i+5} X_i^2 = 0 \ \subset\ \mathbb{P}^4_k. \end{aligned}$$

We therefore mainly consider characteristic $p \geq 5$.

1.1. The main results.

Our first result answers (Q1) affirmatively.

Theorem 1.2 (regular case)**.** *Let k be a field of characteristic p. Let S be a regular projective surface over k with $H^0(S, \mathcal{O}_S) = k$ and $-K_S$ nef. If $p \geq 7$, then S is geometrically integral over k.*

More generally, we prove the following result for log canonical pairs; the same argument yields a precise description of the case $p = 5$.

Theorem 1.3 (general case, = Theorem 5.1)**.** *Let k be a field of characteristic $p \geq 5$. Let S be a projective normal surface over k with $H^0(S, \mathcal{O}_S) = k$, and let Δ be an effective $\mathbb{Q}$-divisor on S such that*

- *$-(K_S + \Delta)$ is nef, and*
- *the pair (S, Δ) is log canonical.*

Then the following hold.

(1) *If $p \geq 7$, then S is geometrically reduced.*

(2) *If $p = 5$ and S is geometrically non-reduced, then $\Delta = 0$, $K_S \equiv 0$, and S has at most canonical singularities. Moreover, writing $\widetilde{S} \to S$ for the minimal resolution, the surface $X := (\widetilde{S} \otimes_k k^{1/p})^{\nu}_{\mathrm{red}}$ has a unique non-canonical singular point P, a rational singularity of multiplicity 3 whose minimal resolution has exceptional locus $E_1 \cup E_2$ with $(E_1)^2_{k^{1/p}} = -3$ and $(E_2)^2_{k^{1/p}} = -2$.*

The bound $p \geq 7$ in Theorem 1.3 (1) is optimal. Guided by the description in Theorem 1.3 (2), and with the help of an AI system, we construct a counterexample in characteristic 5; see §6.1.

Theorem 1.4 (= Theorem 6.1)**.** *Let $k := \mathbb{F}_5(s_1, s_2)$, which has characteristic 5 and degree of imperfection 2. Then there exists a projective* regular *surface S over k with $H^0(S, \mathcal{O}_S) = k$ and $K_S \equiv 0$ such that S is geometrically non-reduced.*

Remark 1.5. (1) Without the log canonical hypothesis, Theorem 1.3 fails in every characteristic (Example 6.3).

(2) Concerning (Q2), we exhibit for $p = 5$ and $p = 7$ regular, geometrically integral, geometrically non-normal surfaces S with $K_S \equiv 0$ (Example 6.2); so $N_2 \geq 7$ if it exists at all.

(3) We show in Proposition 3.1 that, if S admits a smooth Mori fiber space structure over a curve, then S is geometrically regular for all $p \geq 5$.

1.2. **Applications.** Let X be a smooth projective variety over an algebraically closed field of characteristic $p > 0$ with $-K_X$ nef. By [EP23, Theorem 1.3], the Albanese morphism $a_X \colon X \to A$ is surjective; if $X \xrightarrow{f} Y \xrightarrow{g} A$ is its Stein factorization, then $Y \to A$ is purely inseparable, and if moreover f is separable—equivalently, the generic fiber X_η is geometrically integral—then $Y = A$, i.e., a_X is a fibration. When a_X has relative dimension one, it is always a fibration by [CWZ26a, Theorem 1.5]. Combining these with Theorem 1.2 gives the following.

Corollary 1.6 (special case of Corollary 5.2)**.** *Let k be a perfect field of characteristic $p \geq 7$. Let X be a smooth projective variety over k with $-K_X$ nef, and denote by $a_X \colon X \to A$ the Albanese morphism. If $\dim X - \dim a_X(X) \leq 2$, then a_X is a fibration.*

Remark 1.7. In the companion paper [CWZ26b], joint with J. Chen, we establish the effectivity of canonical divisors on regular K-trivial surfaces over arbitrary fields. A key ingredient there is Theorem 1.2: when $p \geq 7$, it is used to show that a regular, geometrically non-normal K-trivial surface S satisfies $K_S \sim 0$.

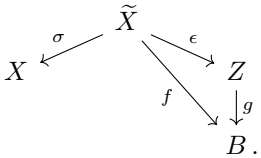
1.3. **Strategy of the proof of Theorem 1.3.** Passing to a minimal resolution, we may assume that S is regular. Since $-K_S \equiv \Delta - (K_S + \Delta)$ is pseudo-effective, either K_S is not pseudo-effective and a K_S-MMP terminates with a Mori fiber space $Z \to B$, or $K_S \equiv 0$. It is not difficult to treat the first case, precisely, it is either a direct consequence of the result of del Pezzo surface or can be proved by similar argument from [PW22] or [BT22]; see §3. The really difficult case from our point of view is $K_S \equiv 0$ which is treated in §4. Let us explain our strategy and the key ingredients to treat this case as follows. We can assume that S is geometrically non-reduced, then $S_{k^{1/p}}$ is non-reduced. Set $X := (S_{k^{1/p}})^{\nu}_{\mathrm{red}}$ and consider the natural morphism $\pi \colon X \to S$. By the formula of Ji–Waldron [JW21, Theorem 1.1]:

$$K_X + (p-1)\mathfrak{C} \equiv 0, \qquad \mathfrak{C} = \mathfrak{M} + \mathfrak{F}, \quad \mathfrak{M} > 0 \text{ movable}, \quad \mathfrak{F} \geq 0. \tag{C1}$$

We stop after this single Frobenius base change because geometric non-reducedness is already visible at $k^{1/p}$; but iterating up to $\overline{k}$ would lose the numerical information we need. More precisely, regularity of S yields

$$pC \text{ is Cartier for every integral curve } C \subset X, \tag{C2}$$

and thus $C \cdot_{k_C} D \in \frac{1}{p}\mathbb{Z}$ for every Weil divisor D, where $k_C := H^0(C^\nu, \mathcal{O}_{C^\nu})$; we refer to this as p-factoriality. Take a minimal resolution $\sigma \colon \widetilde{X} \to X$ and run a $K_{\widetilde{X}}$-MMP, ending with a Mori fiber space $Z \to B$:



Then $K_{\widetilde{X}} + (p-1)(\widetilde{\mathfrak{M}} + \widetilde{\mathfrak{F}}) + \sum_i m_i E_i \equiv 0$ with the E_i the σ-exceptional divisors, and pushing down to Z gives $K_Z + (p-1)(M_Z + F_Z) + E_Z \equiv 0$. For $\dim B = 0$ we invoke boundedness of regular del Pezzo

surfaces [Tan24, Theorem 1.8]. For $\dim B = 1$, since $p-1>2$, one deduces that $\widetilde{\mathfrak{M}}+\widetilde{\mathfrak{F}}$ is g-vertical and that exactly one component of $\sum_i m_i E_i$, say E_1, is horizontal over B; the contradiction then comes from the precise configuration of the E_i and the fibers F_b. Unlike the del Pezzo case, $\widetilde{X}$ may genuinely contain (-1)-curves, as the example of Theorem 1.4 shows, so the MMP is a real birational modification and the situation is much more involved. The following observations, which are consequences of p-factoriality, are what make the configuration tractable:

- a general member $M \in |\mathfrak{M}|$ contains a component $\overline{F}_b := \sigma_* F_b$ for some closed fiber F_b of f over $b \in B$, and we have $(\overline{F}_b)^2_{k(b)} = 1/p$ or $2/p$.
- if $(\overline{F}_b)^2_{k(b)} = 1/p$, then for every closed fiber F_t one has $(\overline{F}_t \cdot \overline{F}_b)_{k(t)} = 1/p$, which forces exactly one component T of F_t, of multiplicity one, to pass through $P = \sigma(E_1)$.

Here we use the notation $(C_1 \cdot C_2)_{k'}$ for the intersection number counted over the field k'.

For $p = 5$ this argument ceases to give a contradiction, and Theorem 1.4 shows that the configuration $(\overline{F}_b)^2_{k(b)} = 2/p$ is genuinely realized.

Remark 1.8. When S is geometrically integral but geometrically non-normal, the linear system $|\mathfrak{C}|$ usually has no movable part, and its strict transform $\widetilde{\mathfrak{C}}$ may be contracted by ϵ. To prove that S is geometrically normal when p is large, the above strategy alone seems not enough, even for $\dim B = 0$.

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AI Statement. An agentic AI system, mainly Fable 5, was used to help construct the characteristic-5 example of Theorem 1.4.

2. Preliminaries

2.1. Conventions.

- Throughout this paper, k is a field of characteristic $p > 0$. By a k-scheme we mean a separated scheme of finite type over k. By a *variety* over k we mean an integral k-scheme. By a *surface* over k we mean a variety of dimension two.
- We write $k \hookrightarrow k^{1/p}$ for the Frobenius map of k, and if X is a scheme over k, then we write $X_{k^{1/p}}$ for $X \times_k k^{1/p}$.
- For a scheme X, we denote by X_{red} the associated reduced scheme and by X^{ν}_{red} its normalization.
- For a variety X over k, we denote by k_X the algebraic closure of k in the function field $K(X)$. Since X is of finite type over k, k_X is a finite extension of k.
- For a birational morphism $\sigma\colon X \to Y$ of varieties and a Weil divisor D on Y, we write $\sigma_*^{-1}(D)$ for the strict transform of D on X.
- By a *fibration* $f\colon X \to B$ we mean a proper surjective morphism to a normal variety with $f_*\mathcal{O}_X = \mathcal{O}_B$.
- Let X be a regular projective surface with $H^0(X, \mathcal{O}_X) = k$. We call X a *del Pezzo surface* if $-K_X$ is ample. We call a fibration $f\colon X \to B$ a *Mori fiber space* if $\dim B < \dim X$, $\rho(X/B) = 1$, and $-K_X$ is relatively ample over B.
- By a *pair* (X, Δ) we mean a normal variety X and an effective $\mathbb{Q}$-divisor Δ such that $K_X + \Delta$ is $\mathbb{Q}$-Cartier. We refer to [Kol13, Definition 2.8] for the definitions of canonical and log canonical singularities.

2.2. Field extension and singularities.

Let X be a variety over k.

(1) If X is proper and normal, then $H^0(X, \mathcal{O}_X) = k_X$ ([Tan21, Proposition 2.1]).

(2) The variety X is geometrically irreducible if and only if $K(X) \cap k^s = k$, where k^s is the separable closure of k ([Liu02, Corollary 3.2.14]). In particular, if k is algebraically closed in $K(X)$, then X is geometrically irreducible.

(3) Combining (1) and (2), we see that if X is a normal proper variety with $k_X = k$, then X is geometrically integral if and only if it is geometrically reduced.

(4) If X and Y are varieties over k birational to each other, then X is geometrically reduced over k if and only if Y is geometrically reduced over k [Tan21, Proposition 2.12].
(5) If X is regular, then for any separable field extension $k \subset k'$ the base change $X \times_k k'$ is regular ([Tan21, Lemma 2.6]).
(6) The variety X is geometrically regular (resp. geometrically reduced, geometrically normal) if and only if $X \times_k k^{1/p}$ is regular (resp. reduced, normal) ([Tan21, Proposition 2.10]).

2.3. **Foliation theory.** We refer to [RS78, Eke87, MP97] for the basic theory of foliations. Since the ground field k is imperfect, we work with subsheaves of the absolute tangent sheaf T_{X/k_0} rather than of the relative tangent sheaf $T_{X/k}$.

Definition 2.1. Let k_0 be a perfect field of characteristic p, let k/k_0 be a finitely generated field extension, and let X be a normal variety over k. A *foliation* on X is an $\mathcal{O}_X$-submodule $\mathcal{F} \subseteq T_{X/k_0}$ that is saturated (i.e. $T_{X/k_0}/\mathcal{F}$ is torsion-free), involutive, and p-closed.

For a foliation $\mathcal{F}$ on X, the sheaf $\mathcal{O}_X^{\mathcal{F}}$ of $\mathcal{F}$-invariant functions is defined by

$$\mathcal{O}_X^{\mathcal{F}}(U) = \big\{ f \in \mathcal{O}_X(U) \;\big|\; \delta(f) = 0 \text{ for all } \delta \in \mathcal{F}(U) \big\}.$$

The *quotient* by $\mathcal{F}$ is the normal variety

$$X/\mathcal{F} := \operatorname{Spec}_X \mathcal{O}_X^{\mathcal{F}},$$

together with the finite purely inseparable morphism $\pi\colon X \to X/\mathcal{F}$ of degree $p^{\operatorname{rk}\mathcal{F}}$. Since the derivations in $\mathcal{F}$ are only k_0-linear, $\mathcal{O}_X^{\mathcal{F}}$ need not be a sheaf of k-algebras. Hence, the quotient $X/\mathcal{F}$ is naturally just a variety over the subfield

$$k^{\mathcal{F}} := \big\{ a \in k \;\big|\; \delta(a) = 0 \text{ for all local sections } \delta \text{ of } \mathcal{F} \big\} \subseteq k.$$

When X is regular, the quotient $X/\mathcal{F}$ is regular at $\pi(x)$ if and only if $\mathcal{F}$ is a subbundle of T_{X/k_0} at x, i.e. the stalk $(T_{X/k_0}/\mathcal{F})_x$ is a free $\mathcal{O}_{X,x}$-module [Eke87, Proposition 2.4]. In particular, if $\mathcal{F}$ is a subbundle everywhere, then $X/\mathcal{F}$ is regular, and the canonical bundle formula of [Eke87, Corollary 3.4] reads

$$\omega_X \cong \pi^* \omega_{X/\mathcal{F}} \otimes (\det \mathcal{F})^{\otimes(p-1)}. \tag{2.1}$$

More generally, let X be a normal variety. Then $\det \mathcal{F} := (\bigwedge^{\operatorname{rk}\mathcal{F}} \mathcal{F})^{\vee\vee}$ is a rank-1 reflexive sheaf and hence determines a Weil divisor class $c_1(\det \mathcal{F})$. Restricting to the big open subset on which X is regular and $\mathcal{F}$ is a subbundle, and then extending Weil divisor classes across codimension at least two, the formula (2.1) yields

$$K_X \sim \pi^* K_{X/\mathcal{F}} + (p-1)\, c_1(\det \mathcal{F}). \tag{2.2}$$

The following example shows that a quotient by a foliation can have a smaller field of constants and be regular but geometrically nonreduced over that field.

Example 2.2. Let k_0 be a perfect field of characteristic $p > 0$ and let $\ell = k_0(s,t)$, so that ℓ has p-degree 2. Set $k = \ell(a,b)$ with $a^p = s$ and $b^p = t$. Let $X = \mathbb{P}^1_k$ with affine coordinate u, and let

$$\delta = \partial_a - u\,\partial_b \in \operatorname{Der}_{k_0}\big(K(X)\big).$$

Then $\delta^{[p]} = 0$, and

$$\mathcal{F} := T_{X/k_0} \cap K(X) \cdot \delta$$

is a saturated rank-1 foliation. On constants, $\delta(a) = 1$ and $\delta(b) = -u$. For $f \in k = \ell(a,b)$, we have $\delta(f) = f_a - u f_b$. Thus $\delta(f) = 0$ implies $f_a = f_b = 0$, so $k^{\mathcal{F}} = \ell$. Let

$$C := \big\{ x^p + s y^p + t z^p = 0 \big\} \subset \mathbb{P}^2_\ell.$$

Its function field is $K(C) = \operatorname{Frac}\big(\ell[\xi,\eta]/(\xi^p + s\eta^p + t)\big)$, where $\xi = x/z$ and $\eta = y/z$ on $C \cap \{z \neq 0\}$. The morphism

$$\phi\colon X \to C, \qquad [U : V] \mapsto [-aU - bV : U : V]$$

is finite purely inseparable of degree p. On the chart $X \cap \{V \neq 0\}$, we have $\phi^*\xi = -au - b$ and $\phi^*\eta = u$. Hence $\phi^* K(C) = \ell(u, au + b) = \ker \delta$. Thus $\mathcal{F} = \ker(T_{X/k_0} \to \phi^* T_{C/k_0})$ and therefore $X/\mathcal{F} \simeq C$ as varieties over $\ell = k^{\mathcal{F}}$. Clearly, the curve C is regular and geometrically nonreduced over $H^0(C, \mathcal{O}_C) = \ell$.

2.4. **Intersection theory.** Let X be a proper variety over k. Let D be a Cartier divisor on X, and C an integral curve on X. We define a 0-cycle on C as follows

$$(D \cdot C)_{\mathrm{cycle}} := \sum \mathrm{ord}_{\mathfrak{p}_i}(D|_C)\mathfrak{p}_i = \sum n_i \mathfrak{p}_i.$$

For the definition of $\mathrm{ord}_{\mathfrak{p}_i}(D|_C)$, we refer the reader to [Ful98, §1.2] or [Liu02, §7.2.1]; for instance, if $\mathfrak{p} \in C$ is a regular closed point and $r \in \mathcal{O}_{C,\mathfrak{p}}$ a local equation for D, then

$$\mathrm{ord}_{\mathfrak{p}}(D) = \max\{n \mid r \in \mathfrak{m}_{C,\mathfrak{p}}^n\}.$$

Definition 2.3 (Intersection number)**.** With the above notation, we define

$$(D \cdot C)_k := \deg_k(\sum n_i \mathfrak{p}_i) = \sum n_i [\kappa(\mathfrak{p}_i) : k].$$

This definition extends to $\mathbb{Q}$- or $\mathbb{R}$-Cartier divisors D, and to 1-cycles C, but in this subsection we keep D Cartier and C an integral curve.

Remark 2.4. The intersection number $(D \cdot C)_k$ coincides with $\deg_k(D|_C)$ as defined in [Liu02, Definition 7.3.1]. By the Riemann–Roch theorem (cf. [Liu02, Theorem 7.3.17]), this number can be equivalently expressed as

$$(D \cdot C)_k = \chi_k(C, D|_C) - \chi_k(C, \mathcal{O}_C).$$

Here for a proper variety X over a field k and a coherent sheaf M on X, we use the convention

$$\chi_k(X, M) := \sum_{i=0}^{\dim X} (-1)^i \dim_k H^i(X, M).$$

Proposition 2.5. *With notation as above, let $\nu\colon C^\nu \to C$ be the normalization of C. Then*

$$(D \cdot C)_k = \sum_{\mathfrak{q}} \mathrm{ord}_{\mathfrak{q}}(D|_{C^\nu})[\kappa(\mathfrak{q}) : k],$$

where the sum runs over the closed points $\mathfrak{q}$ of C^ν.

Proof. See [Ful98, Example 1.2.3]. □

Sometimes, it is convenient to count over the field $\ell := H^0(C^\nu, \mathcal{O}_{C^\nu})$ and write

$$(D \cdot C)_\ell := \sum_{\mathfrak{q}} \mathrm{ord}_{\mathfrak{q}}(D|_{C^\nu})[\kappa(\mathfrak{q}) : \ell] = \frac{1}{[\ell : k]}(D \cdot C)_k.$$

We will also denote this intersection number by $D \cdot_\ell C$. For self-intersections, we use the notation $(C)_k^2 := (C \cdot C)_k$.

2.5. **Adjunction formula.**

Proposition 2.6 ([Kol13, Proposition 2.35])**.** *Let X be a normal surface and $C \subset X$ a reduced curve, and let $C^\nu \to C$ be its normalization. Assume that $K_X + C$ is $\mathbb{Q}$-Cartier.*

(1) *There exists an effective divisor Δ_{C^ν} on C^ν such that*

$$(K_X + C)|_{C^\nu} \sim_{\mathbb{Q}} K_{C^\nu} + \Delta_{C^\nu}.$$

More precisely, if $n(K_X + C)$ is Cartier for some positive integer n, then $n\Delta_{C^\nu}$ is a $\mathbb{Z}$-divisor and

$$n(K_X + C)|_{C^\nu} \sim nK_{C^\nu} + n\Delta_{C^\nu}.$$

(2) *$\Delta_{C^\nu} = 0$ if and only if C is regular and X is regular along C.*

Example 2.7. If X is a normal projective surface that is regular along a regular integral curve $C \subset X$ such that $p_a(C) = 0$, then we have

$$(K_X + C) \cdot_{k_C} C = \deg_{k_C}(K_C) = -2.$$

2.6. **Behavior of canonical divisors under Frobenius base change.** The original statement of the following result assumes that k is F-finite, but this assumption can be dropped by a standard argument (see [DW22, p. 3917]).

Theorem 2.8 ([JW21, Theorem 1.1])**.** *Let X be a normal projective variety over k with $H^0(X, \mathcal{O}_X) = k$. Set $Y := (X \times_k k^{1/p})^\nu_{\mathrm{red}}$ and denote by $\pi\colon Y \to X$ the induced morphism. Then there exists an effective divisor $\mathfrak{C}$ with movable part $\mathfrak{M}$ and fixed part $\mathfrak{F}$ on Y such that*

$$\pi^* K_X \sim K_Y + (p-1)(\mathfrak{M} + \mathfrak{F}).$$

Furthermore, $\mathfrak{C} > 0$ if and only if X is geometrically non-normal, and $\mathfrak{M} > 0$ if and only if X is geometrically non-reduced.

2.7. **Blow-up.** Let X be a surface and let $x \in X$ be a regular closed point. Denote by $\mathcal{I} \subseteq \mathcal{O}_X$ the ideal sheaf corresponding to the closed point $x \in X$. The blow-up of X at x is

$$\mu : \widetilde{X} = \mathrm{Bl}_x(X) := \mathrm{Proj} \bigoplus_{i \geq 0} \mathcal{I}^i \to X.$$

The exceptional curve $E \cong \mathrm{Proj} \bigoplus_i \mathfrak{m}_x^i/\mathfrak{m}_x^{i+1}$ satisfies $H^0(E, \mathcal{O}_E) = k(x)$, and the variety $\widetilde{X}$ is regular along E (cf. [Liu25, Proposition 3.1]). For an effective Cartier divisor D on X, we have

$$\mu^* D = \widetilde{D} + \mathrm{ord}_x(D)E, \tag{2.3}$$

where $\widetilde{D}$ is the strict transform of D (cf. [Ful98, Example 4.3.9]). We remind the reader that even when X is smooth over k, the blow-up $\widetilde{X}$ need not be smooth over k and may even be geometrically non-normal; see [BFSZ24, Remark 2.7].

2.8. **Contractions and exceptional curves.** Let $(x \in X)$ be a germ of a normal surface singularity. Let $\mu\colon \widetilde{X} \to X$ be the minimal resolution of the singularity, and let $E = \bigcup E_i$ be the reduced exceptional locus. We say that $(x \in X)$ is a *rational singularity* if $\dim_k(R^1\mu_*\mathcal{O}_{\widetilde{X}})_x = 0$.

Recall from [Art66, pp. 131–132] that there is a unique nonzero cycle $Z = \sum r_i E_i$ ($r_i \in \mathbb{Z}_{\geq 0}$) that is *minimal* among all nonzero cycles $Z' = \sum a_i E_i$ ($a_i \in \mathbb{Z}_{\geq 0}$) satisfying $(Z' \cdot E_i) \leq 0$ for all i. We call Z the *fundamental cycle* of $\bigcup E_i$.

Proposition 2.9 (Castelnuovo's contraction criterion, [Lip69, Theorem 27.1])**.** *Let X be a normal projective surface over k. Let $E_1, \ldots, E_n$ be distinct integral curves on X such that $\bigcup_i E_i$ is connected and the intersection matrix $((E_i \cdot_k E_j))$ is negative-definite. Let Z be the fundamental cycle of $\bigcup_i E_i$. Then there exists $h\colon X \to Y$ contracting $\bigcup_i E_i$ rationally to a point P if and only if $\chi(Z) > 0$. When this condition holds, Y is regular at P if and only if the* multiplicity $m_P := -(Z)_k^2/h^0(Z, \mathcal{O}_Z)$ *of P on Y is* 1.

Definition 2.10. Let X be a normal projective surface over k. An integral curve $C \subset X$ is called an *exceptional curve of the first kind* (or simply a (-1)*-curve*) if X is regular along C and $K_X \cdot_{k_C} C = C \cdot_{k_C} C = -1$.

Remark 2.11. If $C \subset X$ is a (-1)-curve, then it follows from Proposition 2.9 that C is contractible to a regular point. Moreover, we have $C \cong \mathbb{P}^1_{k_C}$ by [Kol13, Lemma 10.8 (4)].

The following lemma appears in the proof of [Ume81, Theorem 1].

Lemma 2.12. *Let $\sigma\colon X \to Y$ be a composite of (-1)-curve contractions between regular projective surfaces, and let $D \geq 0$ be a divisor on X with $K_X + D \equiv 0$. Then* $\mathrm{Supp}(D)$ *and* $\mathrm{Supp}(\sigma_* D)$ *have the same number of connected components.*

Proof. We may assume that $\sigma\colon X \to Y$ is a contraction of a single (-1)-curve E. Since $K_X + D \equiv 0$, we have $K_Y + \sigma_* D \equiv 0$, and therefore $K_X + D \equiv \sigma^*(K_Y + \sigma_* D)$. Write $D = \mu E + R$ with $\mu \geq 0$ and $E \not\subset \mathrm{Supp}(R)$. Substituting $\sigma^* K_Y \equiv K_X - E$ and $\sigma^*\sigma_* R \equiv R + (R \cdot_{k_E} E)E$ into $K_X + D \equiv \sigma^*(K_Y + \sigma_* D)$ yields $\mu = R \cdot_{k_E} E - 1$. If $\mu = 0$, then $R \cdot_{k_E} E = 1$, so E meets $\mathrm{Supp}(R)$ in exactly one point, and contracting E preserves the number of connected components. If $\mu > 0$, then $E \subset \mathrm{Supp}(D)$, so contracting E again preserves the number of connected components. □

2.9. **Del Pezzo surfaces.**

Theorem 2.13 (MMP for regular surfaces, [Tan18, p. 5] and [Tan20, Theorem 1.2])**.** *Let X be a regular projective surface. Then we can run a K_X-MMP that terminates with either a regular good minimal model or a Mori fiber space $f\colon Z \to B$.*

Theorem 2.14 ([PW22, Theorem 1.5] and [BT22, Proposition 5.2])**.** *If $p \geq 5$, then a regular (or more generally normal and Gorenstein) del Pezzo surface is geometrically normal. If $p \geq 11$, then a regular del Pezzo surface X is geometrically regular.*

Theorem 2.15 ([Tan24, Theorems 1.1 and 1.8])**.** *If X is a regular del Pezzo surface, then the linear system $|-12K_X|$ is very ample over k. Moreover,*

- *if $p \geq 5$ or X is geometrically reduced, then $K_X^2 \leq 9$;*
- *if $p = 3$, then $K_X^2 \leq \max\{9, 3^{\epsilon(X/k)+1}\}$, where $\epsilon(X/k)$ is the thickening exponent introduced in* [Tan21]*;*
- *if $p = 2$, then $K_X^2 \leq \max\{9, 2^{\epsilon(X/k)+3}\}$.*

2.10. **Mori fiber spaces.** Throughout this subsection, unless otherwise stated, $\pi\colon X \to C$ is a Mori fiber space from a regular projective surface X over a field k with $H^0(C, \mathcal{O}_C) = k$.

Proposition 2.16 ([BT22, Proposition 2.18])**.** *If $\dim C = 1$, then for every (not necessarily closed) point $b \in C$ the fiber X_b is an integral conic in $\mathbb{P}^2_{\kappa(b)}$ with $H^0(X_b, \mathcal{O}_{X_b}) = \kappa(b)$. Moreover, if $p \geq 3$ and k is separably closed, then π is a smooth morphism.*

Lemma 2.17. *Let $f\colon X \to C$ be a smooth Mori fiber space from a regular surface X to a curve C with $H^0(C, \mathcal{O}_C) = k$. Then X is geometrically regular over k if and only if C is geometrically regular over k.*

Proof. Consider the Frobenius base change $f_{k^{1/p}}\colon X_{k^{1/p}} \to C_{k^{1/p}}$:

$$\begin{array}{ccc} X_{k^{1/p}} & \xrightarrow{\sigma'} & X \\ {\scriptstyle f_{k^{1/p}}}\downarrow & & \downarrow{\scriptstyle f} \\ C_{k^{1/p}} & \xrightarrow{\sigma} & C. \end{array}$$

Smoothness and surjectivity are preserved by base change, so $f_{k^{1/p}}$ is smooth and faithfully flat. Therefore [Mat89, Theorem 23.7] shows that $X_{k^{1/p}}$ is regular if and only if $C_{k^{1/p}}$ is regular. The claim follows. □

When $\dim C = 1$, we have $\rho(X) = 2$, and $\overline{\mathrm{NE}}(X)$ has exactly two extremal rays. Every vertical curve Γ_v satisfies $\Gamma_v \equiv \mu\, \pi^*(b_0)$ for some $\mu > 0$, so the vertical classes span one of these rays; we let $\mathfrak{f} \in \overline{\mathrm{NE}}(X)$ be its generator normalized by $-K_X \cdot_k \mathfrak{f} = 2$. Equivalently, $\mathfrak{f} = \frac{1}{[k(b_0):k]}\, \pi^*(b_0)$ for one (and hence any) closed point $b_0 \in C$. The class $\mathfrak{f}$ is nef with $\mathfrak{f}^2 = 0$, and $\mathfrak{f} \cdot_k \Gamma = [K(\Gamma) : K(C)]$ for every horizontal integral curve $\Gamma \subset X$.

Lemma 2.18. *Let $f\colon X \to C$ be a smooth Mori fiber space from a regular projective surface onto a curve with $H^0(C, \mathcal{O}_C) = k$. Then $(K_{X/C})^2 = 0$; consequently $(K_X)^2 = -4 \deg_k K_C$.*

Proof. The generic fiber X_η is a smooth conic, so it becomes isomorphic to $\mathbb{P}^1$ over some finite separable extension $L/K(C)$. Let $h\colon C' \to C$ be the normalization of C in L and form

$$\begin{array}{ccc} X' := X \times_C C' & \xrightarrow{\phi} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ C' & \xrightarrow{h} & C. \end{array}$$

The L-point of the generic fiber extends uniquely to a section of f'. A smooth conic fibration with a section is a $\mathbb{P}^1$-bundle; hence there is a rank-2 vector bundle $\mathcal{E}'$ on C' such that $X' \cong \mathbb{P}_{C'}(\mathcal{E}')$. By [Har77, Exercise III.8.4(b)] we have $\omega_{X'/C'} \cong \mathcal{O}_{X'}(-2) \otimes f'^* \det \mathcal{E}'$, and $\xi := c_1(\mathcal{O}_{X'}(1))$ satisfies $\xi^2 = f'^* c_1(\mathcal{E}') \cdot \xi$ by the Grothendieck relation (see [Har77, §A.3]); hence

$$(K_{X'/C'})^2 = (-2\xi + f'^* c_1(\mathcal{E}'))^2 = 4\xi^2 - 4\,\xi \cdot f'^* c_1(\mathcal{E}') = 0.$$

Since ϕ is the base change of h, we have $K_{X'/C'} \sim \phi^* K_{X/C}$, so $\delta\,(K_{X/C})^2 = (K_{X'/C'})^2 = 0$, where $\delta := \deg \phi = [L : K(C)]$. Since $f^* K_C \equiv (\deg_k K_C)\, \mathfrak{f}$ and $K_X \cdot_k \mathfrak{f} = -2$, we conclude

$$(K_X)^2 = (K_{X/C})^2 + 2\, K_{X/C} \cdot_k f^* K_C = -4 \deg_k K_C. \qquad \square$$

The following lemma is a key step in the proof of the main theorem.

Lemma 2.19. *Let $f\colon Y \to B$ be a fibration from a regular projective surface to a regular curve. Suppose that f factors as*

$$Y \xrightarrow{\epsilon} Z \xrightarrow{g} B,$$

where ϵ is a composition of contractions of (-1)-curves and g is a Mori fiber space. Let F_{t_0} be a reducible fiber over some closed point $t_0 \in B$. Suppose that F_{t_0} contains a unique (-1)-curve T and denote $k_T := H^0(T^\nu, \mathcal{O}_{T^\nu})$. Then either $\mathrm{mult}_T(F_{t_0}) \geq 2$, or $\mathrm{mult}_T(F_{t_0}) = 1$ and $[k_T : k(t_0)] > 1$.

Proof. By Proposition 2.16, the fiber $G_{t_0} := g^* t_0$ is an integral conic over $\kappa := k(t_0)$ with $H^0(G_{t_0}, \mathcal{O}_{G_{t_0}}) = \kappa$. Decompose ϵ as

$$Y = Y_N \xrightarrow{\epsilon_N} Y_{N-1} \xrightarrow{\epsilon_{N-1}} \cdots \xrightarrow{\epsilon_2} Y_1 \xrightarrow{\epsilon_1} Y_0 = Z,$$

where each $\epsilon_i\colon Y_i \to Y_{i-1}$ is the blow-up of a regular closed point $P_i \in Y_{i-1}$. Since blow-ups centered outside the fiber over t_0 do not affect F_{t_0}, we may assume that every P_i lies over t_0. Let $E_i \subset Y_i$ be the exceptional curve of ϵ_i. We also denote its strict transform on Y_j by E_i for every $j \geq i$.

Suppose for contradiction that F_{t_0} has a unique (-1)-curve T with $\mathrm{mult}_T(F_{t_0}) = 1$ and $k_T = \kappa$. Then $E_N = T$. Moreover, uniqueness forces $P_i \in E_{i-1} \subset Y_{i-1}$ for every $2 \le i \le N$; otherwise E_{i-1} and E_i would be disjoint (-1)-curves on Y_i, and subsequent blow-ups would leave at least one (-1)-curve above each of them on Y. Hence

$$\kappa \subseteq k(P_1) \subseteq \cdots \subseteq k(P_N) = k_T = \kappa,$$

so every P_i is κ-rational. Set $m_i = \mathrm{mult}_{E_i}(F_{t_0})$. Then

$$1 \le m_1 \le \cdots \le m_N = \mathrm{mult}_T(F_{t_0}) = 1.$$

Hence P_1 is a regular point of multiplicity one on G_{t_0}, so $G_{t_0} \simeq \mathbb{P}^1_\kappa$. Since $(G_{t_0})^2_\kappa = 0$, the first blow-up produces two (-1)-curves of multiplicity one: E_1 and the strict transform of G_{t_0}. If $N = 1$, this already contradicts the uniqueness of T. Thus $N \ge 2$, and P_2 must be the intersection of E_1 and the strict transform of G_{t_0}; otherwise Y would contain at least two (-1)-curves. But then E_2 has multiplicity 2, contradicting $m_2 = 1$. □

3. Geometric regularity of Mori fiber spaces

Proposition 3.1. *Let k be a field of characteristic $p \ge 5$. Let $f\colon X \to C$ be a smooth Mori fiber space from a regular surface X onto a curve C with $H^0(C, \mathcal{O}_C) = k$. Suppose that there exists an effective $\mathbb{Q}$-divisor Δ such that*

(1) *$-(K_X + \Delta)$ is nef, and*

(2) *for every component Γ of Δ that maps birationally to C, we have $\mathrm{coeff}_\Gamma(\Delta) \le 1$.*

Then X is geometrically regular.

Proof. Assume for contradiction that X is geometrically non-regular. By Lemma 2.17, the curve C is geometrically non-regular, and hence geometrically non-normal. Since $p \ge 5$, we have $\deg_k K_C > 0$ by Theorem 2.8. By Lemma 2.18, we have $(K_{X/C})^2 = 0$ and thus $(K_X)^2 = -4 \deg_k K_C < 0$. Since $-(K_X + \Delta)$ is nef, $\Delta^2 \le (-(K_X + \Delta) + \Delta)^2 = (K_X)^2 < 0$. Hence there is a component Γ of Δ such that $(\Gamma)^2_k < 0$. Since all fibers of f are irreducible, this implies that Γ dominates C.

We can write that $\Delta = c\Gamma + \Delta'$ where $c = \mathrm{coeff}_\Gamma(\Delta)$. Let $\nu\colon \Gamma^\nu \to \Gamma$ be the normalization, set $k_\Gamma := H^0(\Gamma^\nu, \mathcal{O}_{\Gamma^\nu})$ and denote $\psi := f \circ \nu\colon \Gamma^\nu \to C$. Set $e_\Gamma := [K(\Gamma) : K(C)]$. Then $e_\Gamma = \Gamma \cdot_k \mathfrak{f}$, where $\mathfrak{f} = \frac{1}{[k(b_0):k]} f^*(b_0)$ as in Subsection 2.10. Since $-(K_X + \Delta)$ is nef, $-(K_X + \Delta) \cdot_k \mathfrak{f} = 2 - ce_\Gamma - \Delta' \cdot_k \mathfrak{f} \ge 0$, and it follows that

$$c\, e_\Gamma \le 2 - \Delta' \cdot_k \mathfrak{f} \le 2. \tag{3.1}$$

Considering the intersection with Γ, we have

$$(K_X + c\,\Gamma) \cdot_{k_\Gamma} \Gamma \le (K_X + c\,\Gamma + \Delta') \cdot_{k_\Gamma} \Gamma = (K_X + \Delta) \cdot_{k_\Gamma} \Gamma \le 0. \tag{3.2}$$

In turn, applying the Adjunction formula (Proposition 2.6) we obtain

$$0 \ge (K_X + c\,\Gamma) \cdot_{k_\Gamma} \Gamma = (K_X + \Gamma) \cdot_{k_\Gamma} \Gamma + (c-1)(\Gamma)^2_{k_\Gamma} \ge \deg_{k_\Gamma} K_{\Gamma^\nu} + (c-1)(\Gamma)^2_{k_\Gamma}. \tag{3.3}$$

If the projection $\Gamma \to C$ is separable, then by Riemann–Hurwitz [Liu02, Theorem 7.4.16], we have $\deg_k K_{\Gamma^\nu} > 0$. Remark that $c \le 1$ always holds: this is obvious when $e_\Gamma \ge 2$ by (3.1); and when $e_\Gamma = 1$ this is assumption (2). We derive a contradiction by the inequality (3.3) since $(\Gamma)^2_{k_\Gamma} < 0$.

Now assume $\Gamma \to C$ is inseparable, which implies that $p \mid e_\Gamma$. Since $X \to C$ is a Mori fiber space, we can take the basis $K_X, \mathfrak{f}$ of $\mathrm{NS}(X)_\mathbb{Q}$. Combining the information $(\mathfrak{f})^2_k = 0$, $K_X \cdot_k \mathfrak{f} = -2$ and $\Gamma \cdot_k \mathfrak{f} = e_\Gamma$, we can write that

$$\Gamma \equiv -\frac{e_\Gamma}{2} K_X + b\mathfrak{f}$$

for some $b \in \mathbb{Q}$. Set $\deg_k K_C = d$. Then $(K_X)^2_k = -4d$. Notice that $\deg_{k_\Gamma}(f^* K_C|_\Gamma) = \frac{de_\Gamma}{[k_\Gamma:k]} \in \mathbb{Z}$; this implies $[k_\Gamma : k] \mid de_\Gamma$. Considering the intersection numbers with Γ over k_Γ, we have,

$$K_X \cdot_{k_\Gamma} \Gamma = \frac{K_X \cdot_k \Gamma}{[k_\Gamma : k]} = \frac{2(de_\Gamma - b)}{[k_\Gamma : k]}, \quad (\Gamma)^2_{k_\Gamma} = \frac{-e_\Gamma(de_\Gamma - 2b)}{[k_\Gamma : k]} = -\frac{e_\Gamma}{2}(K_X \cdot_{k_\Gamma} \Gamma) + \frac{be_\Gamma}{[k_\Gamma : k]} < 0.$$

Since the above numbers are integers, $2b$ is also divisible by $[k_\Gamma : k]$. By the inequality (3.2) we have

$$(K_X + c\,\Gamma) \cdot_{k_\Gamma} \Gamma = \frac{2(de_\Gamma - b) - ce_\Gamma(de_\Gamma - 2b)}{[k_\Gamma : k]} \ge \frac{2(de_\Gamma - b) - 2(de_\Gamma - 2b)}{[k_\Gamma : k]} = \frac{2b}{[k_\Gamma : k]} \le 0,$$

and thus $K_X \cdot_{k_\Gamma} \Gamma > 0$. In turn, we conclude that

(∗) $-(\Gamma)^2_{k_\Gamma} \ge \frac{e_\Gamma}{2}(K_X \cdot_{k_\Gamma} \Gamma)$; and if moreover $K_X \cdot_{k_\Gamma} \Gamma = 1$ and e_Γ is odd, say, $e_\Gamma = p \ge 5$ then $b < 0$, thus $-(\Gamma)^2_{k_\Gamma} \ge \frac{e_\Gamma}{2} - \frac{be_\Gamma}{[k_\Gamma:k]} \ge e_\Gamma$.

Combining $\deg_{k_\Gamma} K_{\Gamma^\nu} \geq -2$, the inequality (3.3) and $ce_\Gamma \leq 2$, we have

$$0 \geq \deg_{k_\Gamma} K_{\Gamma^\nu} + (c-1)(\Gamma)^2_{k_\Gamma} \geq -2 + (c-1)(\Gamma)^2_{k_\Gamma} \geq -2 + (\frac{2}{e_\Gamma} - 1)(\Gamma)^2_{k_\Gamma},$$

thus

$$(1 - \frac{2}{e_\Gamma})(-(\Gamma)^2_{k_\Gamma}) \leq 2.$$

Then applying the statement $(*)$, we obtain that

$$(1 - \frac{2}{e_\Gamma})(-(\Gamma)^2_{k_\Gamma}) \geq (\frac{e_\Gamma}{2} - 1)(K_X \cdot_{k_\Gamma} \Gamma) \leq 2,$$

this requires $e_\Gamma \leq 5$ and $K_X \cdot_{k_\Gamma} \Gamma = 1$, but then $-(\Gamma)^2_{k_\Gamma} \geq e_\Gamma$, which causes a contradiction by

$$(1 - \frac{2}{e_\Gamma})(-(\Gamma)^2_{k_\Gamma}) \geq e_\Gamma - 2 \geq 3. \qquad \square$$

In Proposition 3.1, both assumption (2) and the smoothness assumption on f are necessary (see Examples 6.3 and 6.4, respectively).

4. Geometric reducedness of K-trivial surfaces

This section is devoted to proving the following result.

Theorem 4.1. *Let ℓ be a field of characteristic $p \geq 5$ and let $k := \ell^{1/p}$ be the Frobenius extension of ℓ. Let S be a projective regular surface over ℓ such that $H^0(S, \mathcal{O}_S) = \ell$ and $K_S \equiv 0$. Assume that S is geometrically non-reduced.*

Then $p = 5$, and $X := (S \otimes_\ell k)^\nu_{\mathrm{red}}$ has a unique non-canonical singular point P, which is a rational singularity of multiplicity 3, and whose minimal resolution has exceptional locus $E_1 \cup E_2$ with $(E_1)^2_k = -3$, $(E_2)^2_k = -2$.

Proof. By the properties in Subsection 2.2, $S_{\ell^{\mathrm{sep}}}$ is regular, and we only need to prove the statements for $S_{\ell^{\mathrm{sep}}}$. So we may assume that ℓ is separably closed. By assumption, $S \times_\ell k$ is not reduced. Set $X = (S \times_\ell k)^\nu_{\mathrm{red}}$ and denote by $\pi \colon X \to S$ the induced morphism. By Theorem 2.8, we have

$$0 \equiv \pi^* K_S \sim K_X + (p-1)(\mathfrak{M} + \mathfrak{F}), \tag{4.1}$$

where $\mathfrak{M} \neq 0$ is movable and $\mathfrak{F}$ is effective. Let $\sigma \colon \widetilde{X} \to X$ be a minimal resolution. We have

$$\sigma^* K_X = K_{\widetilde{X}} + \sum_i a_i E_i,$$

where $a_i \geq 0$ and E_i's are σ-exceptional curves. Then $\kappa(\widetilde{X}) < 0$, and therefore a $K_{\widetilde{X}}$-MMP yields a birational contraction $\epsilon \colon \widetilde{X} \to Z$, obtained by contracting (-1)-curves, onto a Mori fiber space $g \colon Z \to B$:

$$\begin{array}{ccccc} & & \widetilde{X} & & \\ & \swarrow^{\sigma} & & \searrow^{\epsilon} & \\ S & \xleftarrow{\pi} & X = (S \times_\ell k)^\nu_{\mathrm{red}} & & Z \\ \downarrow & & \downarrow & \swarrow_{f} & \downarrow{\scriptstyle g} \\ \operatorname{Spec} \ell & \longleftarrow & \operatorname{Spec} k & & B\,. \end{array} \tag{4.2}$$

Recall that, for a variety Y over k, the field k_Y denotes the algebraic closure of k in $K(Y)$. In the present situation, we have $k_X = k_{\widetilde{X}} = k_B = k$.

We shall take advantage of the following two conditions:

(a) Since S is regular and $X \to S$ is of height one, the Cartier index of any Weil divisor on X divides p. In particular, for every irreducible divisor C on X, if we write $\sigma^* C = \widetilde{C} + \sum_i e_i E_i$, where $\widetilde{C}$ denotes the strict transform of C, then $e_i \in \frac{1}{p}\mathbb{Z}_{\geq 0}$; and for any two distinct irreducible divisors C_1 and C_2, if $C_1 \cdot_k C_2 > 0$, then $C_1 \cdot_{k_{C_1}} C_2 \geq \frac{1}{p}$ (which is equivalent to $C_1 \cdot_k C_2 \geq \frac{1}{p}[k_{C_1} : k]$).

(b) We write $\sigma^*(K_X) = K_{\widetilde{X}} + \sum_i a_i E_i$, and $\sigma^*(\mathfrak{M} + \mathfrak{F}) = \widetilde{\mathfrak{M}} + \widetilde{\mathfrak{F}} + \sum_i b_i E_i$, where $\widetilde{\mathfrak{M}} = \sigma_*^{-1}\mathfrak{M}$, $\widetilde{\mathfrak{F}} = \sigma_*^{-1}\mathfrak{F}$, and $a_i, b_i \in \frac{1}{p}\mathbb{Z}_{\geq 0}$. Then

$$K_{\widetilde{X}} + (p-1)(\widetilde{\mathfrak{M}} + \widetilde{\mathfrak{F}}) + \sum_i m_i E_i \equiv 0 \tag{4.3}$$

where $m_i := a_i + (p-1)b_i$. Since K_S is Cartier, $m_i \in \mathbb{Z}_{\geq 0}$.

Lemma 4.2. *The Mori fiber space $Z \to B$ is a fibration onto a curve B.*

Proof. Suppose to the contrary that Z is a del Pezzo surface with $\rho(Z) = 1$. Pushing down (4.1), we have

$$K_Z + \epsilon_*\big((p-1)\sigma^*(\mathfrak{M}+\mathfrak{F}) + \sum_i a_i E_i\big) \equiv 0.$$

Since $\mathfrak{M}$ is movable, the divisor $\epsilon_*(\sigma^*\mathfrak{M}) > 0$ is nonzero and effective, and thus ample. Consequently,

$$(-K_Z)_k^2 \geq \big(\epsilon_*((p-1)\sigma^*\mathfrak{M})\big)_k^2 \geq (p-1)^2 \geq 16,$$

which contradicts that $(K_Z)_k^2 \leq 9$ (see Theorem 2.15). □

Lemma 4.3. *Each component of $\widetilde{\mathfrak{M}}+\widetilde{\mathfrak{F}}$ is contained in a fiber of f, and the divisor $\sum_i m_i E_i$ dominates B.*

Proof. Let $\eta \in B$ be the generic point. By the condition (b), we have

$$K_{\widetilde{X}_\eta} + \big((p-1)(\widetilde{\mathfrak{M}}+\widetilde{\mathfrak{F}}) + \sum_i m_i E_i\big)\big|_{\widetilde{X}_\eta} \equiv 0.$$

Since $\deg_{k(\eta)} K_{\widetilde{X}_\eta} = -2$ and $p \geq 5$, we conclude that $(\widetilde{\mathfrak{M}}+\widetilde{\mathfrak{F}})|_{\widetilde{X}_\eta} = 0$; that is, every component of $\widetilde{\mathfrak{M}}+\widetilde{\mathfrak{F}}$ is vertical, and it follows that $\sum_i m_i E_i$ dominates B. □

For a closed point $b \in B$, denote by $F_b = f^*b$ the fiber of f over b, by $k(b)$ the residue field of b, and set $\overline{F}_b = \sigma_* F_b$. A general fiber of f is integral and normal, with $H^0(F_b, \mathcal{O}_{F_b}) = k(b)$. Since $\mathfrak{M}$ is movable, the strict transform of a general member $M_0 \in |\mathfrak{M}|$ contains a fiber F_b. Therefore,

(c) there exist a reduced and normal fiber F_b and an effective Weil divisor $\overline{V}$ such that $\mathfrak{M}+\mathfrak{F} = \overline{F}_b + \overline{V}$.

We fix such a fiber F_b and the divisor $\overline{V}$, and denote by V the strict transform of $\overline{V}$. Let E_1 be a σ-exceptional curve that dominates B; it exists by Lemma 4.3 and we have $m_1 > 0$.

Lemma 4.4. *Among the σ-exceptional curves with $m_i > 0$, E_1 is the unique one dominating B. Moreover, $F_b \cdot_{k(b)} E_1 = 1$, $m_1 = 2$, $F_b \cdot E_i = 0$ for every $i \neq 1$ with $m_i > 0$, and $F_b \cdot V = 0$. Consequently $k_{E_1} = k$, E_1 is a section of f (hence is regular), and V and every E_i with $i \neq 1$ and $m_i > 0$ are vertical.*

Proof. By restricting the relation $K_{\widetilde{X}} + (p-1)(F_b+V) + \sum_i m_i E_i \equiv 0$ to F_b and applying the adjunction formula, we obtain

$$K_{F_b} + \Big((p-2)F_b + (p-1)V + m_1 E_1 + \sum_{i\geq 2} m_i E_i\Big)\Big|_{F_b} \equiv 0.$$

By taking the degree and using the facts that $F_b \cdot_{k(b)} F_b = 0$ and $\deg_{k(b)} K_{F_b} = -2$, we obtain

$$m_1 E_1 \cdot_{k(b)} F_b + (p-1)V \cdot_{k(b)} F_b + \sum_{i\geq 2} m_i E_i \cdot_{k(b)} F_b = 2. \tag{4.4}$$

By restricting the same relation to E_1 we obtain

$$K_{E_1^\nu} + \Delta_{E_1^\nu} + \Big((m_1-1)E_1 + (p-1)F_b + (p-1)V + \sum_{i\geq 2} m_i E_i\Big)\Big|_{E_1} \equiv 0,$$

where $\Delta_{E_1^\nu} \geq 0$ is the boundary on E_1 obtained from the adjunction formula. Taking degree gives

$$(m_1-1)(E_1)^2_{k_{E_1}} + (p-1)F_b \cdot_{k_{E_1}} E_1 + (p-1)V \cdot_{k_{E_1}} E_1 + \sum_{i\geq 2} m_i E_i \cdot_{k_{E_1}} E_1 \leq 2. \tag{4.5}$$

Since $(p-1)F_b \cdot_{k_{E_1}} E_1 \geq 4$, we conclude from (4.5) that $m_1 \geq 2$. Combining this with (4.4), we have

$$m_1 = 2, \quad E_1 \cdot_{k(b)} F_b = 1, \quad \text{and} \quad V \cdot F_b = \sum_{i\geq 2} m_i E_i \cdot_{k(b)} F_b = 0. \tag{4.6}$$

Moreover, E_1 is the unique σ-exceptional curve horizontal over B since $K_{\widetilde{X}} \cdot_{k(b)} F_b = -2$. Every E_i (for $i \geq 2$ and $m_i > 0$) is therefore vertical, hence disjoint from the general fiber F_b, which gives $F_b \cdot E_i = 0$. Finally, since $E_1 \cdot_{k(b)} F_b = 1$, the curve E_1 is a section of $f\colon \widetilde{X} \to B$; in particular $k_{E_1} = k$ and E_1 is regular. □

Let $P = \sigma(E_1) \in X$ and denote by $E_1, \ldots, E_r$ the exceptional curves centered at P. Write $\sigma^*\overline{F}_b = F_b + \sum_i c_i E_i$ and $\sigma^*\overline{V} = V + \sum_i d_i E_i$, where $c_i, d_i \in \frac{1}{p}\mathbb{Z}_{\geq 0}$ by (a). Then $m_i = a_i + (p-1)(c_i+d_i)$, with a_i as in (b).

Lemma 4.5. (1) *$c_i > 0$ for all $1 \le i \le r$; and*

(2) *$a_1 = 2/p$ or $(p+1)/p$.*

Proof. (1) Intersecting the pullback $\sigma^*\overline{F}_b$ with E_j gives

$$0 = \sigma^*\overline{F}_b \cdot_k E_j = F_b \cdot_k E_j + \sum_{i=1}^{r} c_i\,(E_i \cdot_k E_j), \qquad 1 \le j \le r. \tag{4.7}$$

For $j = 1$, since $F_b \cdot_k E_1 > 0$ and $(E_1)_k^2 < 0$, we must have $c_1 > 0$. Suppose $T := \{\, i : c_i = 0 \,\} \neq \varnothing$. Since $\bigcup_i E_i$ is connected and $1 \notin T$, there exist $j_0 \in T$ and $i_0 \notin T$ with $E_{i_0} \cdot_k E_{j_0} > 0$. Evaluating (4.7) at j_0 then forces $c_{i_0}(E_{i_0} \cdot_k E_{j_0}) \le 0$, a contradiction. Hence $c_i > 0$ for all i.

(2) Since $2 = m_1 = a_1 + (p-1)(c_1 + d_1)$, we see that $(c_1 + d_1) = 1/p$ or $2/p$, and the statement follows. □

Lemma 4.6. *If $a_1 = 2/p$, then we have $p = 5$, $k(b) = k_{E_1} = k$, $p_a(E_1) = 0$, $\deg_k K_{E_1} = -2$, $F_b \cdot_k E_1 = 1$, $V \cdot_k E_1 = 0$, and $(E_1)_k^2 \in \{-2, -3\}$.*

Proof. By restricting the relation $\sigma^*(K_X) = K_{\widetilde{X}} + a_1E_1 + \sum_{i\ge2} a_iE_i$ to E_1 and applying the adjunction formula, we obtain

$$K_{E_1} + \big((a_1 - 1)E_1 + \sum_{i\ge2} a_iE_i\big)\big|_{E_1} \equiv 0.$$

Taking the degree gives

$$\deg_k(K_{E_1}) + (a_1 - 1)(E_1)_k^2 + \sum_{i\ge2} a_iE_i \cdot_k E_1 = 0. \tag{4.8}$$

We deduce that

$$(E_1)_k^2 = \frac{p}{p-2}\big(\deg_k K_{E_1} + \sum_{i\ge2} a_iE_i \cdot_k E_1\big) \ge \frac{p}{p-2}(\deg_k K_{E_1}) \ge \frac{-2p}{p-2}.$$

Since $(E_1)^2 < 0$, we have $\deg_k K_{E_1} < 0$, hence $p_a(E_1) = 0$ and $\deg_k K_{E_1} = -2$. Moreover, since σ is a minimal resolution, $(E_1)_k^2 \le -2$, hence

$$(E_1)_k^2 = \begin{cases} -3 \text{ or } -2 & \text{if } p = 5, \\ -2 & \text{if } p \ge 7. \end{cases} \tag{4.9}$$

Furthermore, by Lemma 4.4, the inequality (4.5) can be rewritten as

$$(E_1)_k^2 + (p-1)F_b \cdot_k E_1 + (p-1)V \cdot_k E_1 + \sum_{i\ge2} m_iE_i \cdot_k E_1 \le 2. \tag{4.10}$$

Combining (4.10) with (4.9) and $F_b \cdot_k E_1 \ge 1$, we conclude that $p = 5$, $F_b \cdot_k E_1 = 1$, and $V \cdot_k E_1 = 0$. Finally, $F_b \cdot_k E_1 = 1$ implies $k(b) = k$. □

Lemma 4.7. (1) *$\overline{V} \cdot \overline{F}_b = 0$;*

(2) *$\overline{F}_b \cdot_{k(b)} \overline{F}_b = 1/p$ or $2/p$; and*

(3) *$V \cdot_k E_i = 0$ for $i = 1, \dots, r$ (in particular, $d_i = 0$ for $i = 1, \dots, r$).*

Proof. As before, write $\sigma^*\overline{F}_b = F_b + c_1E_1 + \sum_{i\ge2} c_iE_i$ and $\sigma^*\overline{V} = V + d_1E_1 + \sum_{i\ge2} d_iE_i$. Then

$$\begin{aligned} \overline{F}_b \cdot_{k(b)} \overline{V} &= F_b \cdot_{k(b)} \sigma^*\overline{V} = d_1, \quad \text{and} \\ (\overline{F}_b)_{k(b)}^2 &= \sigma^*\overline{F}_b \cdot_{k(b)} F_b = c_1. \end{aligned} \tag{4.11}$$

(1) Assume $\overline{V} \cdot \overline{F}_b = d_1 > 0$ for a contradiction. Since $c_1 > 0$ (Lemma 4.5 (1)) and $2 = m_1 = a_1 + (p-1)(c_1 + d_1)$, we have $c_1 = d_1 = 1/p$ and $a_1 = 2/p$. By Lemma 4.6, $p = 5$, $F_b \cdot_k E_1 = 1$ and $(E_1)_k^2 \in \{-2, -3\}$. Intersecting $\sigma^*\overline{F}_b$ with E_1 now gives

$$0 = \sigma^*\overline{F}_b \cdot_k E_1 = F_b \cdot_k E_1 + c_1(E_1)_k^2 + \sum_{i\ge2} c_i\,E_i \cdot_k E_1 \ge 1 - \tfrac{3}{5} > 0,$$

a contradiction. Therefore $\overline{V} \cdot \overline{F}_b = 0$.

(2) By (4.11) and (1), we have $(\overline{F}_b)_{k(b)}^2 = c_1$ and $d_1 = 0$, so $2 = m_1 = a_1 + (p-1)c_1$. Since $a_1 \ge 0$, $c_1 \in \frac{1}{p}\mathbb{Z}_{>0}$, and $3(p-1) > 2p$ for $p \ge 5$, we obtain $(\overline{F}_b)_{k(b)}^2 = c_1 = 1/p$ or $2/p$.

(3) Since $\overline{V} \cdot_k \overline{F}_b = 0$ and $F_b \cdot_k V = 0$, we have

$$0 = \overline{F}_b \cdot_k \overline{V} = \sigma^*\overline{F}_b \cdot_k V = F_b \cdot_k V + \sum_i c_i (E_i \cdot_k V) = \sum_i c_i (E_i \cdot_k V).$$

Each term is nonnegative and $c_i > 0$ by Lemma 4.5 (1), so $V \cdot_k E_i = 0$ for $i = 1, \dots, r$. Since $\sigma^*\overline{V} = V + \sum_{i\geq 1} d_i E_i$, for each $j = 1, \dots, r$ we have $0 = \sum_{i=1}^r d_i E_i \cdot_k E_j$, whence $d_i = 0$ by the negative definiteness of the intersection matrix of the E_i. □

For convenience, we write

$$K_{\widetilde{X}} + \mathbf{D} \equiv 0, \qquad \mathbf{D} = (p-1)(F_b + V) + E' + E'', \qquad E' = \sum_{i=1}^r m_i E_i, \tag{4.12}$$

where E' is the exceptional divisor centered at P, and E'' is the sum of the exceptional divisors centered away from P.

Lemma 4.8. (1) $\operatorname{Supp} \mathbf{D}$ *is connected.*

(2) $V = 0$ *and* $E'' = 0$*; in particular,* P *is the only non-canonical singularity on* X*.*

Proof. (1) The contraction $\epsilon\colon \widetilde{X} \to Z$ from running the $K_{\widetilde{X}}$-MMP is a composite of (-1)-curve contractions, so by Lemma 2.12 it preserves the number of connected components of $\operatorname{Supp} \mathbf{D}$. Since E_1 is the only horizontal component of $\mathbf{D}$, $\epsilon_*\mathbf{D}$ is the horizontal curve $\epsilon_* E_1$ together with vertical components; since the fibers of $Z \to B$ are irreducible, each such vertical component is a whole fiber meeting $\epsilon_* E_1$. Hence $\operatorname{Supp}(\epsilon_*\mathbf{D})$, and therefore $\operatorname{Supp}(\mathbf{D})$, is connected.

(2) By Lemma 4.7 (3), V is disjoint from E'; moreover V contains no component of the integral fiber F_b (otherwise $\sigma^*\overline{V} \supseteq \sigma^*\overline{F}_b$ would give $d_1 \geq c_1 > 0$), so V, being vertical, is disjoint from F_b. The curves of E'' lie in fibers different from F_b and, being exceptional over points different from P, are disjoint from E'. Hence $V \cup E''$ does not meet $F_b \cup E'$; so (1) forces $V = 0$ and $E'' = 0$. □

By Lemma 4.7 (3), we have $d_1 = 0$, so $(\overline{F}_b)^2_{k(b)} = c_1$ and $2 = m_1 = a_1 + (p-1)c_1$; hence we fall into one of the following two cases:

- Case (1) $(\overline{F}_b)^2_{k(b)} = 2/p \iff a_1 = 2/p$;
- Case (2) $(\overline{F}_b)^2_{k(b)} = 1/p \iff a_1 = (p+1)/p$.

Proposition 4.9. *In Case* (1)*, we have* $p = 5$*, and* X *has a unique non-canonical singular point* P*, which is a rational singularity of multiplicity* 3 *whose minimal resolution has exceptional locus* $E = E_1 + E_2$ *with* $(E_1)^2_k = -3$ *and* $(E_2)^2_k = -2$*.*

Proof. In this case $a_1 = c_1 = 2/p$. From Lemma 4.6 we have $p = 5$, $p_a(E_1) = 0$, $F_b \cdot_k E_1 = 1$, $(E_1)^2_k = -2$ or -3. Intersecting $\sigma^*\overline{F}_b = F_b + \sum_{i=1}^r c_i E_i$ with E_1, we obtain

$$F_b \cdot_k E_1 + \frac{2}{5}(E_1)^2_k + \sum_{i\geq 2} c_i E_i \cdot_k E_1 = 0,$$

and thus

$$-\frac{2}{5}(E_1)^2_k = 1 + \sum_{i\geq 2} c_i E_i \cdot_k E_1 \geq 1.$$

Therefore, $(E_1)^2_k = -3$ and thus $\sum_{i\geq 2} c_i E_i \cdot_k E_1 = 1/p$. Since $c_i \in \frac{1}{p}\mathbb{Z}_{\geq 0}$, there exists exactly one E_i, say E_2, such that $E_2 \cdot_k E_1 > 0$ and $c_2 > 0$, and more precisely $E_2 \cdot_k E_1 = 1$ and $c_2 = 1/p$. Here $E_2 \cdot_k E_1 = 1$ implies $k_{E_2} = k_{E_1} = k$.

Since $m_2 \geq (p-1)c_2 \geq (p-1)/p$ and $m_2 \in \mathbb{Z}$, inequality (4.10) then implies $m_2 = 1$ and $\sum_{i\geq 3} m_i E_i \cdot_k E_1 = 0$. Restricting the same relation to E_2 and using $F_b \cdot_k E_2 = V \cdot_k E_2 = 0$ and $E_1 \cdot_k E_2 = 1$, we obtain

$$\deg_k(K_{E_2^\nu}) + \deg_k(\Delta_{E_2^\nu}) + 2 + \Big(\sum_{i\geq 3} m_i E_i\Big) \cdot_k E_2 = 0, \tag{4.13}$$

where $\Delta_{E_2^\nu} \geq 0$ is the boundary obtained from the adjunction formula along E_2. Since $\deg_k(K_{E_2^\nu}) \geq -2$, it follows that $p_a(E_2^\nu) = 0$, $\Delta_{E_2^\nu} = 0$, and $\sum_{i\geq 3} m_i E_i \cdot_k E_2 = 0$; hence E_2 is regular. Using $\sum_{i\geq 3} m_i E_i \cdot_k E_2 = 0$ and $m_i \geq (p-1)c_i > 0$, we see that $r = 2$ and the exceptional divisor centered at P is $E' = 2E_1 + E_2$. Moreover, by the equation

$$0 = E_2 \cdot_k \sigma^*\overline{F}_b = E_2 \cdot_k (F_b + c_1 E_1 + c_2 E_2) = \frac{2}{p} E_1 \cdot_k E_2 + \frac{1}{p}(E_2)^2_k,$$

we deduce $(E_2)_k^2 = -2$. Finally, since E_1 and E_2 are of arithmetic genus zero with $E_1 \cdot_k E_2 = 1$, P is a rational singularity; it is then straightforward to verify that it has multiplicity 3. □

We are left to deal with Case (2).

Lemma 4.10. *In Case (2), there is some σ-exceptional curve E_2 meeting E_1.*

Proof. Assume E_1 is the only irreducible component of $\sigma^{-1}(P)$. Then

$$0 = \sigma^* \overline{F}_b \cdot_k E_1 = F_b \cdot_k E_1 + c_1(E_1)_k^2,$$

where $F_b \cdot_k E_1 = [k(b) : k]$. Since $c_1 = 1/p$, we have $(E_1)_k^2 = -p[k(b) : k]$. Restricting $K_{\widetilde{X}} + (p-1)F_b + 2E_1$ to the regular curve E_1 gives

$$\deg_k K_{E_1} + (E_1)_k^2 + (p-1)[k(b) : k] = 0.$$

Then $[k(b) : k] = \deg_k K_{E_1}$. Since k is assumed to be separably closed, $[k(b) : k]$ is a power of p; in particular, $[k(b) : k]$ is odd. This contradicts the fact that $\deg_k K_{E_1}$ is even. □

Lemma 4.11. *Assume $(\overline{F}_b)_{k(b)}^2 = 1/p$. Then for every closed point $t \in B$, the fiber F_t has at most one non-σ-exceptional component T with $\overline{T} \cdot_k \overline{F}_b > 0$, and any such T has $\mathrm{mult}_{F_t} T = 1$.*

Proof. From $\overline{F}_t \equiv \frac{[k(t):k]}{[k(b):k]} \overline{F}_b$ and $(\overline{F}_b)_{k(b)}^2 = 1/p$ we get

$$\overline{F}_b \cdot_k \overline{F}_t = \tfrac{1}{p}[k(t) : k].$$

Write $\overline{F}_t = \sum_j e_j \overline{T}_j$, where T_j runs over the non-σ-exceptional components of F_t, with $e_j = \mathrm{mult}_{F_t} T_j \geq 1$. If $\overline{T}_j \cdot_k \overline{F}_b > 0$, then condition (a) gives

$$\overline{T}_j \cdot_k \overline{F}_b \geq \tfrac{1}{p}[k_{T_j} : k] \geq \tfrac{1}{p}[k(t) : k].$$

Since $\frac{1}{p}[k(t) : k] = \overline{F}_b \cdot_k \overline{F}_t = \sum_j e_j \, (\overline{T}_j \cdot_k \overline{F}_b)$ is a sum of nonnegative terms with $e_j \geq 1$, at most one such T_j occurs, and for it $e_j = 1$. □

Proposition 4.12. *Case (2) does not occur.*

Proof. Suppose to the contrary that Case (2) holds, that is, $(\overline{F}_b)_{k(b)}^2 = c_1 = 1/p$. By Lemma 4.8 we have

$$K_{\widetilde{X}} + (p-1)F_b + E' \equiv 0, \qquad E' = \sum_{i=1}^{r} m_i E_i. \tag{4.14}$$

By Lemma 4.10, we have $r \geq 2$. Let F_{t_0} be the fiber of f containing E_2. Then it is reducible and contains at least one (-1)-curve. Let T be a (-1)-curve in F_{t_0}; since σ is a minimal resolution, T is different from every E_i. Intersecting (4.14) with T therefore gives

$$E' \cdot_{k_T} T = -K_{\widetilde{X}} \cdot_{k_T} T = 1.$$

Thus T meets exactly one component E_{i_0} of the exceptional divisor centered at P, with $E_{i_0} \cdot_{k_T} T = 1$. Hence *every* (-1)-curve $T \subseteq F_{t_0}$ satisfies $\sigma_* T \neq 0$ and $\overline{T} \cdot_k \overline{F}_b > 0$. By Lemma 4.11, this implies that F_{t_0} has a unique (-1)-curve, of multiplicity one. Lemma 2.19 then gives $[k_T : k(t_0)] > 1$. On the other hand, the proof of Lemma 4.11 gives

$$\tfrac{1}{p}[k(t_0) : k] \geq \overline{T} \cdot_k \overline{F}_b \geq \tfrac{1}{p}[k_T : k] \geq \tfrac{1}{p}[k(t_0) : k].$$

Thus $[k_T : k] = [k(t_0) : k]$, and hence $k_T = k(t_0)$, a contradiction. □

We are in either Case (1) or Case (2). Proposition 4.12 excludes Case (2), so we are in Case (1), and Proposition 4.9 concludes the proof of Theorem 4.1. □

5. Generalization and Application

Theorem 5.1. *Let k be a field of characteristic $p \geq 5$. Let S be a projective normal surface over k such that $H^0(S, \mathcal{O}_S) = k$. Let Δ be an effective $\mathbb{Q}$-divisor on S and assume that*

- *$-(K_S + \Delta)$ is nef, and*
- *the pair (S, Δ) is log canonical.*

Then the following hold.

(1) *If $p \geq 7$, then S is geometrically integral.*

(2) *If $p = 5$ and S is geometrically non-reduced, then $\Delta = 0$, $K_S \equiv 0$, the surface S has at most canonical singularities, and $X := (\widetilde{S} \otimes_k k^{1/p})^{\nu}_{\mathrm{red}}$, where $\widetilde{S} \to S$ denotes the minimal resolution, has a unique non-canonical singular point P, which is a rational singularity of multiplicity 3, whose minimal resolution has exceptional locus $E_1 \cup E_2$ with $(E_1)^2_{k^{1/p}} = -3$ and $(E_2)^2_{k^{1/p}} = -2$.*

Proof. We may assume that k is separably closed. Assume that S is geometrically non-reduced. Let $\sigma\colon \widetilde{S} \to S$ be a minimal resolution with exceptional divisors E_i, and write

$$\sigma^* K_S = K_{\widetilde{S}} + \sum a_i E_i, \qquad \sigma^* \Delta = \sigma_*^{-1}\Delta + \sum b_i E_i,$$

where $a_i \geq 0$ since σ is a minimal resolution, and $b_i \geq 0$ since Δ is effective and $\mathbb{Q}$-Cartier. Set $\widetilde{\Delta} = \sigma_*^{-1}\Delta + \sum(a_i + b_i)E_i$, so that $K_{\widetilde{S}} + \widetilde{\Delta} = \sigma^*(K_S + \Delta)$ is anti-nef. Since (S, Δ) is log canonical, we have $a_i + b_i \leq 1$, hence $(\widetilde{S}, \widetilde{\Delta})$ is again log canonical. Note that $\widetilde{S}$ is geometrically non-reduced and $H^0(\widetilde{S}, \mathcal{O}_{\widetilde{S}}) = k$. In the following, we discuss two cases: (i) $K_{\widetilde{S}} \not\equiv 0$; (ii) $K_{\widetilde{S}} \equiv 0$.

(i) Assume $K_{\widetilde{S}} \not\equiv 0$. Set $D := -(K_{\widetilde{S}} + \widetilde{\Delta})$, which is nef. Then $-K_{\widetilde{S}} = \widetilde{\Delta} + D$ is pseudo-effective, and thus $K_{\widetilde{S}}$ cannot be pseudo-effective; otherwise $K_{\widetilde{S}} \equiv 0$, contrary to our assumption. Then the $K_{\widetilde{S}}$-MMP (Theorem 2.13) yields a sequence of (-1)-curve contractions $\phi\colon \widetilde{S} \to Z$, followed by a Mori fiber space $f\colon Z \to B$. Set $\Delta_Z := \phi_*\widetilde{\Delta}$ and $D_Z := \phi_* D$. Then $-(K_Z + \Delta_Z) \equiv D_Z$ is nef, and Δ_Z satisfies the same coefficient bounds as $\widetilde{\Delta}$. If $\dim B = 0$, then Z is geometrically normal by Theorem 2.14. If $\dim B = 1$, then f is smooth by Proposition 2.16, so Proposition 3.1 shows that Z is geometrically regular. Thus Z is geometrically normal in either case. Since $\widetilde{S} \to Z$ is birational, the surface $\widetilde{S}$ is geometrically reduced, a contradiction. So Case (i) does not occur.

(ii) Assume $K_{\widetilde{S}} \equiv 0$. Theorem 4.1 shows that $p = 5$ and $X = (\widetilde{S} \otimes_k k^{1/p})^{\nu}_{\mathrm{red}}$ satisfies the stated description. It remains to prove the assertions on (S, Δ) in (2). Since $-(K_{\widetilde{S}} + \widetilde{\Delta}) \equiv -\widetilde{\Delta}$ is nef and $\widetilde{\Delta}$ is effective, we have $\widetilde{\Delta} = 0$. In particular $\Delta = 0$ and $a_i = 0$ for all i, so S has canonical singularities. Moreover $\sigma^* K_S = K_{\widetilde{S}} \equiv 0$, so $K_S \equiv 0$ by the projection formula. □

Corollary 5.2. *Let k be a perfect field of characteristic $p \geq 7$. Let X be a $\mathbb{Q}$-factorial projective variety. Suppose that there exists an effective $\mathbb{Q}$-divisor Δ on X such that (X, Δ) is strongly F-regular and $-(K_X + \Delta)$ is nef with Cartier index not divisible by p. If the Albanese morphism $a_X\colon X \to A_X$ has relative dimension ≤ 2 over its image, then a_X is a fibration.*

Proof. Denote by $X \xrightarrow{h} W \xrightarrow{g} A_X$ the Stein factorization of a_X. Since (X, Δ) is strongly F-regular, it is klt [HW02, Theorem 3.3]. If $\dim X = \dim W$, that is, if X has maximal Albanese dimension, then the conclusion follows from [EP23, Proposition 3.2] or [CWZ26a, Proposition 2.9]. If $\dim X - \dim W = 1$, then the conclusion follows from [CWZ26a, Theorem 1.5]. Finally, assume $\dim X - \dim W = 2$. The generic fiber X_η over $\eta = \operatorname{Spec} K(W)$ is a surface with $H^0(X_\eta, \mathcal{O}_{X_\eta}) = K(W)$, (X_η, Δ_η) is klt, and $-(K_{X_\eta} + \Delta_\eta)$ nef. Since $p \geq 7$, Theorem 5.1(1) shows that X_η is geometrically integral, and hence h is separable. Therefore a_X is a fibration by [EP23, Theorem 1.3]. Note that the assumption on the Cartier index of $-(K_X + \Delta)$ is needed to apply [EP23], and the $\mathbb{Q}$-factoriality of X is needed to apply [CWZ26a]. □

6. Examples

6.1. A geometrically non-reduced K-trivial surface in characteristic 5.

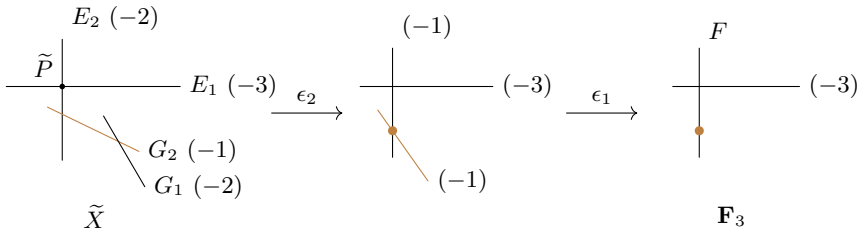
Theorem 6.1. *Let $\ell = \mathbb{F}_5(s_1, s_2)$ be a field of characteristic 5 and degree of imperfection 2. Then there exists a projective regular surface S over ℓ with $H^0(S, \mathcal{O}_S) = \ell$ and $K_S \equiv 0$ such that S is geometrically non-reduced over ℓ.*

Construction idea: Assume that S is such a surface. Set $k := \ell^{1/5}$ and let $\widetilde{X}$ be the minimal resolution of $X := (S \times_\ell k)^{\nu}_{\mathrm{red}}$:

$$\begin{array}{ccccc} \widetilde{X} & \xrightarrow{\sigma} & X & \xrightarrow[/\mathcal{F}]{\pi} & S \\ & & \downarrow & & \downarrow \\ & & \operatorname{Spec} k & \longrightarrow & \operatorname{Spec} \ell. \end{array}$$

Since $[k : \ell] = 5^2$ and $S \times_\ell k$ is non-reduced, we have $\deg(\pi) = 5$ and thus $S \cong X/\mathcal{F}$ for some rank-1 foliation $\mathcal{F}$ on X. According to Proposition 4.9, we can get a candidate for $\widetilde{X}$ by blowing up

$\mathbf{F}_3 = \mathbb{P}(\mathcal{O} \oplus \mathcal{O}(3)) \to \mathbb{P}^1_k$ as follows:



(6.1)

There exists a normal surface $\widetilde{S}$ fitting into the following commutative diagram

$$\begin{array}{ccc} \widetilde{X} & \xrightarrow{\tilde{\pi}} & \widetilde{S} \\ \downarrow{\scriptstyle \sigma} & & \downarrow{\scriptstyle \bar{\sigma}} \\ X & \xrightarrow{\pi} & S \end{array}$$

such that $\bar{\sigma}$ is birational and $\tilde{\pi}$ is a universal homeomorphism. Let $\overline{E}_i = \tilde{\pi}(E_i)$ for $i = 1, 2$. One can show that

$$K_{\widetilde{S}} = \bar{\sigma}^* K_S + 2\overline{E}_1 + 3\overline{E}_2 \quad \text{and} \quad \tilde{\pi}^* \overline{E}_i = E_i \ (i = 1, 2).$$

Let $\mathcal{D}$ be the foliation on $\widetilde{X}$ corresponding to $\tilde{\pi}$, and write $\mathcal{D} \cong \mathcal{O}_{\widetilde{X}}(-\mathfrak{d})$. Since $K_{\widetilde{X}} + 4F + 2E_1 + E_2 \equiv 0$, and $K_{\widetilde{X}} + 4\mathfrak{d} \equiv \tilde{\pi}^* K_{\widetilde{S}}$, we must have $\mathfrak{d} \equiv F + E_1 + E_2$.

From the above analysis, we only need to find a foliation $\mathcal{D}$ on $\widetilde{X}$ such that

(1) $-\det(\mathcal{D}) \equiv F + E_1 + E_2$,

(2) *and for the induced foliation $\mathcal{F}$ on X, $X/\mathcal{F}$ is regular.*

An AI system found such a foliation $\mathcal{D}$.

Proof of Theorem 6.1. We break the argument into the following steps.

Step 1: The toric model $\widetilde{X}$.

Let $N := \mathbb{Z}^2$ be the lattice with standard basis e_1, e_2, and let $\widetilde{X}$ be the toric surface over k whose fan has primitive ray generators, in cyclic order,

$$(1, 0), \quad (0, 1), \quad (-1, -2), \quad (-2, -5), \quad (-1, -3), \quad (0, -1),$$

with corresponding toric divisors $F_0, C_\infty, G_1, G_2, E_2, E_1$; see Figure 1. Let $M := \mathrm{Hom}(N, \mathbb{Z})$ be the dual

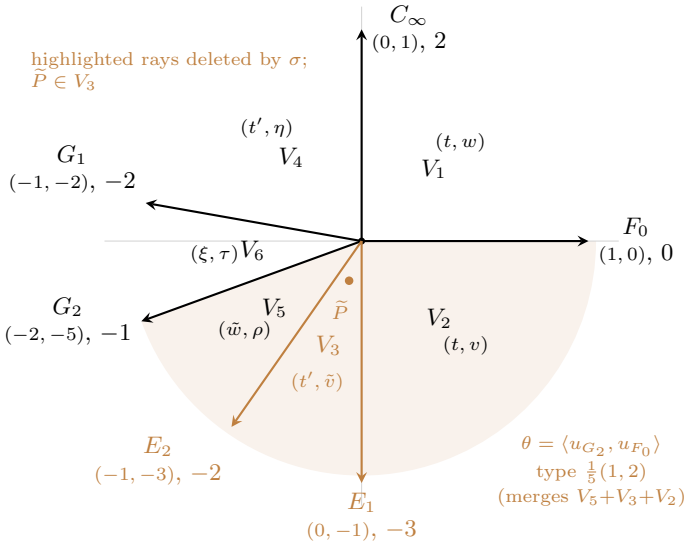


FIGURE 1. The toric surface $\widetilde{X}$

lattice with dual basis $e_1^\vee, e_2^\vee$, and let

$$t := \chi^{e_1^\vee}, \qquad w := \chi^{e_2^\vee}$$

be the corresponding characters of the dense torus. Each consecutive pair has determinant 1, so $\widetilde{X}$ is smooth. Consecutive toric divisors meet once and nonconsecutive ones are disjoint; hence

$$E_1 \cdot_k E_2 = E_2 \cdot_k G_2 = G_2 \cdot_k G_1 = 1,$$

while the remaining pairs among E_1, E_2, G_1, G_2 are disjoint. For a torus-invariant prime divisor D, let u_{prev} and u_{next} be the primitive generators adjacent to u_D. The toric relation

$$u_{\mathrm{prev}} + u_{\mathrm{next}} = -(D)_k^2 u_D$$

gives

$$(E_1)_k^2 = -3, \qquad (E_2)_k^2 = -2, \qquad (G_1)_k^2 = -2, \qquad (G_2)_k^2 = -1.$$

Set $\widetilde{P} := E_1 \cap E_2$.

The rational function t extends to a toric morphism $f\colon \widetilde{X} \to B \cong \mathbb{P}^1_k$. Choose $b \in B(k)$ in the dense torus and put $F_b := f^*(b)$. Then E_1 is a section, E_2 is vertical, $F_b \cdot_k E_1 = 1$, and

$$F := F_b \equiv F_0 \equiv E_2 + G_1 + 2G_2, \tag{6.2}$$

where the last equivalence follows from $\mathrm{div}(t) = F_0 - E_2 - G_1 - 2G_2$. Since the toric boundary is anticanonical, we have

$$-K_{\widetilde{X}} = F_0 + C_\infty + G_1 + G_2 + E_2 + E_1.$$

The relation $\mathrm{div}(w) = C_\infty - E_1 - 2G_1 - 5G_2 - 3E_2$ and (6.2) then give

$$-K_{\widetilde{X}} \equiv F_0 + 2E_1 + 3(E_2 + G_1 + 2G_2) + E_2 \equiv 4F + 2E_1 + E_2. \tag{6.3}$$

Step 2: The foliation $\mathcal{D}$.

Let $c_i = s_i^{1/p}$ $(i = 1, 2)$ so that $k = \ell^{1/p} = \ell(c_1, c_2)$. Write $\partial_{c_1}, \partial_{c_2}$ for the basis of $\mathrm{Der}_\ell(k)$ dual to the p-basis c_1, c_2; their common kernel is ℓ. Set $K := K(\widetilde{X}) = k(t, w)$ and let

$$\delta = \partial_{c_1} + t\,\partial_{c_2} + w\,\partial_t \in \mathrm{Der}_\ell(K).$$

Equivalently, δ is given on the p-basis c_1, c_2, t, w of K over K^5 by

$$\delta(c_1) = 1, \qquad \delta(c_2) = t, \qquad \delta(t) = w, \qquad \delta(w) = 0.$$

Then $\delta^{[5]} = 0$, as one checks on the p-basis, so $K\delta$ is p-closed and $L := \ker \delta$ has index 5 in K. For $\lambda \in k$ one has $\delta(\lambda) = \partial_{c_1}(\lambda) + t\,\partial_{c_2}(\lambda)$. Using the linear independence of $1, t$ over k, together with $\delta(c_1) = 1$ and $[K : L] = 5$, we obtain

$$L \cap k = \ell, \qquad Lk = K. \tag{6.4}$$

Let $\mathcal{D} := T_{\widetilde{X}/\ell} \cap K\delta$ be the saturation of δ. Then $\mathcal{D}$ is a rank-1 foliation on a regular surface $\widetilde{X}$ and thus locally free, so

$$\mathcal{D} \cong \mathcal{O}_{\widetilde{X}}(-\mathfrak{d})$$

for some divisor $\mathfrak{d}$. Put $t' = 1/t$, $v = 1/w$, $\tilde{w} = w/t^3$, $\tilde{v} = 1/\tilde{w}$, $\eta = w/t^2$, $\xi = t^2/w$ and $\rho = t^5/w^2 = 1/\tau$. Clearing denominators and cancelling common factors in the six charts below gives the local generator ∂ of $\mathcal{D}$:

chart	(y_1, y_2)	∂	$\partial(c_1)$	$\partial(c_2)$	$\partial(y_1)$	$\partial(y_2)$
V_1	(t, w)	δ	1	t	w	0
V_2	(t, v)	$v\delta$	v	tv	1	0
V_3	$(t', \tilde{v})$	$t'^2\tilde{v}\,\delta$	$t'^2\tilde{v}$	$t'\tilde{v}$	$-t'$	$3\tilde{v}$
V_4	(t', η)	$t'\delta$	t'	1	$-t'\eta$	$3\eta^2$
V_5	$(\tilde{w}, \rho)$	$\tilde{w}^2\rho^2\delta$	$\tilde{w}^2\rho^2$	ρ	2	0
V_6	(ξ, τ)	$\xi^2\tau\,\delta$	$\xi^2\tau$	1	2	0

For example, in the chart V_5, we have $t = 1/(\tilde{w}^2\rho)$ and $w = 1/(\tilde{w}^5\rho^3)$, so

$$\delta = \partial_{c_1} + \frac{1}{\tilde{w}^2\rho}\,\partial_{c_2} - \frac{3}{\tilde{w}^2\rho^2}\,\partial_{\tilde{w}} + \frac{5}{\tilde{w}^3\rho}\,\partial_\rho.$$

Clearing denominators gives the regular (but not saturated) section $\tilde{w}^3\rho^2\delta$, with coefficients

$$\big(\tilde{w}^3\rho^2,\ \tilde{w}\rho,\ -3\tilde{w},\ 5\rho\big).$$

Then, since $p = 5$, we have $5\rho = 0$, and we can cancel the common factor $\tilde{w}$ to obtain $\tilde{w}^2\rho^2\delta$. One can work out the other charts similarly.

Here $E_1 = \{v = 0\} = \{\tilde{v} = 0\}$; $E_2 = \{t' = 0\}$ in V_3 and $\{\rho = 0\}$ in V_5; $G_1 = \{t' = 0\}$ in V_4 and $\{\tau = 0\}$ in V_6; $G_2 = \{\tilde{w} = 0\}$ in V_5 and $\{\xi = 0\}$ in V_6; and $F_0, C_\infty \subseteq V_1$.

Since $\delta(c_1) = 1$, evaluation at c_1 defines a nonzero section of $\mathcal{D}^\vee \cong \mathcal{O}_{\widetilde{X}}(\mathfrak{d})$. Relative to each local generator ∂ in the table, this section is represented by $\partial(c_1)$. Its vanishing orders along E_1, E_2, G_1, G_2 are respectively $1, 2, 1, 2$, and it has no other zeros. Hence

$$\mathfrak{d} = E_1 + 2E_2 + G_1 + 2G_2. \tag{6.5}$$

Thus by (6.2),

$$\mathfrak{d} \equiv E_1 + E_2 + F, \qquad \mathfrak{d} \cdot_k E_1 = \mathfrak{d} \cdot_k E_2 = -1, \quad \mathfrak{d} \cdot_k F = 1.$$

Moreover, $\partial(v) = 0$ and $\partial(\tilde{v}) = 3\tilde{v}$ show that E_1 is $\mathcal{D}$-invariant, while $\partial(t') = -t'$ and $\partial(\rho) = 0$ show that E_2 is $\mathcal{D}$-invariant; and in each chart except V_3 some value of ∂ is a unit, while in V_3 the four values generate the unit ideal away from $\widetilde{P}$. So $\widetilde{P}$ is the only point at which $\mathcal{D} \subseteq T_{\widetilde{X}/\ell}$ is not a subbundle.

Step 3: Contract and descend.

Deleting the rays u_{E_1}, u_{E_2} defines a projective toric contraction $\sigma\colon \widetilde{X} \to X$; set $P := \sigma(E_1 \cup E_2)$. The germ (X, P) is the affine toric surface of the cone $\langle(-2,-5),(1,0)\rangle$, i.e., the cyclic quotient singularity of type $\frac{1}{5}(1,2)$. Thus X is normal projective, regular away from P, and σ is its minimal resolution. Since $\widetilde{X} \to X$ is birational, $\mathcal{D}$ induces a foliation $\mathcal{F}$ on X; set

$$S := X/\mathcal{F}, \qquad \pi\colon X \to S, \qquad s_0 := \pi(P).$$

Then S is a normal projective surface.

Step 4: Regularity of S.

First consider $x \in \widetilde{X} \setminus (E_1 \cup E_2)$ and let ∂ generate $\mathcal{D}$ at x. By the chart computation in Step 2 there is g with $\partial(g) \in \mathcal{O}^\times_{\widetilde{X},x}$; replacing ∂ by $\partial(g)^{-1}\partial$, which spans the same line, we may assume $\partial(g) = 1$, and then $\partial^{[5]} = c\partial$ with $c = \partial^5(g) = \partial^4(1) = 0$. Hence $A := \ker\partial$ satisfies $\mathcal{O}_{\widetilde{X},x} = \bigoplus_{i=0}^4 Ag^i$, so $A \to \mathcal{O}_{\widetilde{X},x}$ is a faithfully flat local homomorphism and A is regular by [Mat89, Theorem 23.7]. Thus $S \setminus \{s_0\}$ is regular.

We next show that S is regular at s_0. On $V_2 \subset \widetilde{X}$, (t, v) are coordinates and $E_1 = \{v = 0\}$. Set $U := v = 1/w$ and $T := t$. Via the common dense torus of $\widetilde{X}$ and X, these are rational functions on both surfaces, but not local parameters at $P \in X$. On $\widetilde{X}$ one has $\nu_{E_1}(U^aT^b) = a$ and $\nu_{E_2}(U^aT^b) = 3a - b$. The monomial $U^aT^b = t^b w^{-a}$ corresponds to the lattice point $(b, -a) \in M$, hence is regular on the affine chart of X for $\langle(-2,-5),(1,0)\rangle$ if and only if $b \geq 0$ and $5a - 2b \geq 0$. Set $\mathcal{C} := \{(a,b) \in \mathbb{Z}^2 : 0 \leq 2b \leq 5a\}$ and $A := k[U^aT^b : (a,b) \in \mathcal{C}]$, and let $\mathfrak{m} \subset A$ be the maximal ideal corresponding to P. Set

$$R := \mathcal{O}_{X,P} = A_{\mathfrak{m}}.$$

The generator of $\mathcal{D}$ at $\widetilde{P}$ given in Step 2 can be written in terms of U, T as

$$\partial = t'^2\tilde{v}\,\delta = T\partial_T + U\big(T\partial_{c_1} + T^2\partial_{c_2}\big).$$

Thus

$$\partial(\lambda U^aT^b) = \lambda b\, U^aT^b + (\partial_{c_1}\lambda)U^{a+1}T^{b+1} + (\partial_{c_2}\lambda)U^{a+1}T^{b+2} \qquad (\lambda \in k), \tag{6.6}$$

so $\partial(R) \subseteq R$. Since $\partial(U) = 0$ and $\partial(T) = T$, we have $\partial^n(c_1) = UT$ and $\partial^n(c_2) = 2^{n-1}UT^2$ for $n \geq 1$. Thus $2^4 = 1$ implies that $\partial^{[5]}$ and ∂ agree on the generators U, T, c_1, c_2 of $K/\mathbb{F}_5$; as both are derivations in characteristic 5, they are equal. Since $X^5 - X$ is separable, $\partial^{[5]} = \partial$ yields an eigenspace decomposition $R = \bigoplus_{i \in \mathbb{F}_5} R_i$ with $R_i = \{x \in R : \partial(x) = ix\}$; in particular $R_0 = \ker(\partial|_R) = R \cap L = \mathcal{O}_{S,s_0}$, and the projection $R \to R_0$ is $1 - \partial^4$. Filter R by $F_n = \{x : \nu_{E_1}(x) \geq n\}$, so that $\operatorname{gr} R = \bigoplus_n F_n/F_{n+1}$. The formula (6.6) shows $\partial(F_n) \subseteq F_n$ and $(\partial - T\partial_T)(F_n) \subseteq F_{n+1}$. Write $\operatorname{gr}\partial$ for the induced derivation of $\operatorname{gr} R$. Then $\operatorname{gr}\partial = T\partial_T$, with $\operatorname{gr}\partial\,(U^aT^b) = b\,U^aT^b$ in $\operatorname{gr} R$. Hence every element of $\operatorname{gr} R_0$ is a k-linear combination of the initial forms of monomials U^aT^b with $5 \mid b$. Conversely, if $(a,b) \in \mathcal{C}$, $5 \mid b$, and $\lambda \in k$, then the projection $1 - \partial^4 : R \to R_0$ satisfies

$$(1 - \partial^4)(\lambda U^aT^b) = \lambda U^aT^b + (\text{higher } \nu_{E_1}).$$

Thus every such monomial occurs as an initial form in $\operatorname{gr} R_0$. In particular the residue field of R_0 is k. The semigroup $\mathcal{C} \cap \{5 \mid b\}$ is freely generated by $(1,0)$ and $(2,5)$, since $(a, 5m) = (a - 2m)(1,0) + m(2,5)$. Set $u := U$, $r := U^2T^5$, and let $\mathfrak{n}$ be the maximal ideal of R_0. Both u and r lie in R_0. For $0 \neq x \in R_0$ with $d = \nu_{E_1}(x)$, write $\operatorname{in}(x)$ for the image of x in F_d/F_{d+1}. The preceding computation gives

$$\operatorname{gr} R_0 = k[\operatorname{in}(u), \operatorname{in}(r)].$$

For the filtration induced by $F_\bullet$ on $\mathfrak{n}/\mathfrak{n}^2$, this gives

$$\operatorname{gr}(\mathfrak{n}/\mathfrak{n}^2) = k\,\mathrm{in}(u) \oplus k\,\mathrm{in}(r),$$

where the two generators have degrees 1 and 2; all other nonconstant monomials lie in $\mathfrak{n}^2$. Thus u, r generate $\mathfrak{n}/\mathfrak{n}^2$, so Nakayama's lemma gives $\mathfrak{n} = (u, r)$. Since $\dim R_0 = 2$, we obtain

$$R_0 = \mathcal{O}_{S,s_0} \text{ is regular, with regular parameters } \quad u = \frac{1}{w}, \qquad r = \frac{t^5}{w^2}. \tag{6.7}$$

Finally, $U^i(U^2T^5)^j = U^{i+2j}T^{5j}$ has $\nu_{E_1} = i + 2j$ and $\nu_{E_2} = 3i + j$, and distinct (i, j) with equal $i + 2j$ have distinct $3i + j$. So no leading terms cancel and ν_{E_1}, ν_{E_2} are the monomial valuations of weights $(1, 2)$ and $(3, 1)$ in u, r, of discrepancies $1 + 2 - 1 = 2$ and $3 + 1 - 1 = 3$.

Step 5: The conclusion.

Let $\tilde{\pi}\colon \widetilde{X} \to \widetilde{S} := \widetilde{X}/\mathcal{D}$ and $\overline{E}_i := \tilde{\pi}(E_i)$. Since E_i is $\mathcal{D}$-invariant, we have $\tilde{\pi}^*\overline{E}_i = E_i$ by [RS78, Proposition 1]. By (2.2), (6.3) and (6.5),

$$\tilde{\pi}^*K_{\widetilde{S}} \equiv K_{\widetilde{X}} + 4\mathfrak{d} \equiv -(4F + 2E_1 + E_2) + 4(F + E_1 + E_2) = 2E_1 + 3E_2 = \tilde{\pi}^*(2\overline{E}_1 + 3\overline{E}_2),$$

so $K_{\widetilde{S}} \equiv 2\overline{E}_1 + 3\overline{E}_2$ since $\tilde{\pi}^*$ is injective on numerical classes. Since the discrepancies of $\overline{E}_1, \overline{E}_2$ over the regular point s_0 are 2 and 3 (Step 4), the contraction $\widetilde{S} \to S$ of $\overline{E}_1 + \overline{E}_2$ satisfies $K_{\widetilde{S}} = \bar{\sigma}^*K_S + 2\overline{E}_1 + 3\overline{E}_2$, whence $K_S \equiv 0$.

Since $\mathcal{O}_S \subseteq \mathcal{O}_X$, (6.4) gives $H^0(S, \mathcal{O}_S) = k \cap L = \ell$. The same relation yields $[K : L] = [Lk : L] = 5 < [k : \ell]$. By Mac Lane's criterion, a finitely generated extension F/ℓ is separable if and only if $[F \cdot \ell^{1/p} : F] = [\ell^{1/p} : \ell]$. Thus L/ℓ is inseparable, and therefore S is geometrically non-reduced over ℓ. □

6.2. Geometrically non-normal K-trivial surfaces in characteristic ≤ 7.

Example 6.2. Let k_0 be an algebraically closed field of characteristic $p \leq 7$. Let $G = \mu_p$ be the infinitesimal group scheme over k_0. Let C be a smooth curve over k_0 with a free G-action (e.g., C is an ordinary elliptic curve or $C = \mathbb{A}^1 \setminus \{0\}$). By [Mat23, Theorem 7.3 (2), Examples 10.2, 10.3, 10.7 and 10.8], there exists a G-quotient $\rho\colon T \to S$ such that

- S is an RDP K3 surface, i.e., S has at most rational double point singularities and its minimal resolution is a K3 surface;
- T is a proper integral non-normal Gorenstein surface satisfying $\omega_T \cong \mathcal{O}_T$, $\operatorname{Sing}(T) \cap \rho^{-1}(\operatorname{Sing}(S)) = \varnothing$, and $\operatorname{Supp}\operatorname{Fix}(G) = \rho^{-1}(\operatorname{Sing}(S))$.

Let G act diagonally on $T \times C$. Since G acts freely on C, it acts freely on $T \times C$ and thus the quotient morphism

$$q\colon T \times C \longrightarrow W := (T \times C)/G$$

is a finite faithfully flat G-torsor. The projections induce morphisms

$$\begin{array}{ccc} W = (T \times C)/G & \xrightarrow{f} & C/G = C' \\ \downarrow{\scriptstyle g} & & \\ T/G = S\,. & & \end{array}$$

Set $U := S^{\mathrm{reg}}$ and $T_U := \rho^{-1}(U)$. Since $T_U \to U$ is a G-torsor, the associated bundle

$$g^{-1}(U) = (T_U \times C)/G \longrightarrow U$$

is fppf locally the smooth projection $T_U \times C \to T_U$; hence it is smooth and $g^{-1}(U)$ is regular. For $w \in g^{-1}(S \setminus U)$, choose $(t, c) \in T \times C$ above w. Since $\rho(t) \in \operatorname{Sing}(S)$, the stated property of T shows that T is regular at t, so $T \times C$ is regular at (t, c). Regularity descends along q by [Sta25, Tag 07NG], so W is regular at w and hence everywhere.

Set $K := K(C')$ and $L := K(C)$, and let $X := W_K$ be the generic fiber of f. The morphism f is flat and projective. Since W is regular, so is X. By construction, $X_L \cong T_L$, so X is geometrically integral and geometrically non-normal. Moreover, $H^0(X, \mathcal{O}_X) = K$. Finally, since

$$\omega_X|_{X_L} \cong \omega_{X_L} \cong (\omega_T)_L \cong \mathcal{O}_{X_L}$$

and $\operatorname{Pic}(X) \to \operatorname{Pic}(X_L)$ is injective by [Sta25, Tag 0CC5], we have $\omega_X \cong \mathcal{O}_X$. Thus X is a projective, regular, geometrically integral, geometrically non-normal, K-trivial surface.

6.3. **Counterexamples for Mori fiber spaces.** The following example shows that the coefficient bound in Proposition 3.1 is necessary. For each prime p, we construct a geometrically non-reduced Mori fiber space $f\colon X \to C$ with an effective $\mathbb{Q}$-divisor Δ such that (X, Δ) is not log canonical.

Example 6.3. Let $k = \mathbb{F}_p(s,t)$ and let

$$C = (X_0^p + sX_1^p + tX_2^p = 0) \subseteq \mathbb{P}^2_k.$$

Then C is a regular, geometrically non-reduced projective curve with $H^0(C, \mathcal{O}_C) = k$. For $n > 0$, set $\mathcal{E} = \mathcal{O}_C \oplus \mathcal{O}_C(-n)$ and $X = \mathbb{P}_C(\mathcal{E})$. Then $\deg_k \mathcal{E} = -np$, and the projection $f\colon X \to C$ is a smooth Mori fiber space; moreover, X is geometrically non-reduced. Let C_0 be the section corresponding to $\mathcal{E} \twoheadrightarrow \mathcal{O}_C(-n)$ [Har77, Proposition II.7.12]. Then $\mathcal{O}_X(C_0) \cong \mathcal{O}_X(1)$, and the relative Euler sequence gives

$$\omega_X \cong \mathcal{O}_X(-2C_0) \otimes f^*(\omega_C \otimes \det \mathcal{E}).$$

Hence

$$K_X + 2C_0 + (np - \deg_k \omega_C)\mathfrak{f} \equiv 0, \qquad \mathfrak{f} := \frac{f^*(b)}{[k(b) : k]}$$

for any closed point $b \in C$. Choose n with $np \geq \deg_k \omega_C$ and set

$$\Delta := 2C_0 + (np - \deg_k \omega_C)\mathfrak{f}.$$

Then $\Delta \geq 0$ and $-(K_X + \Delta) \equiv 0$, while $\operatorname{coeff}_{C_0}(\Delta) = 2$. Thus (X, Δ) is not log canonical.

The following example shows that the smoothness assumption in Proposition 3.1 is also necessary.

Example 6.4. Let $p > 2$, let a, d be algebraically independent over $\mathbb{F}_p$, and set $k := \mathbb{F}_p(a,d)$ and $C := \mathbb{P}^1_k$ with homogeneous coordinates $[U : V]$. Let $\mathcal{E} := \mathcal{O}_C(-p)^{\oplus 2} \oplus \mathcal{O}_C$, write $\pi\colon \mathbf{P} := \mathbb{P}_C(\mathcal{E}) \to C$, and let x, y, z be the tautological coordinates associated with the three summands, so that x, y are sections of $\mathcal{O}_{\mathbf{P}}(1) \otimes \pi^*\mathcal{O}_C(p)$ and z is a section of $\mathcal{O}_{\mathbf{P}}(1)$. Let $G := (U^p - aV^p)(U^p - UV^{p-1}) \in H^0(C, \mathcal{O}_C(2p))$. Then $x^2 - dy^2 - Gz^2$ is a section of $\mathcal{O}_{\mathbf{P}}(2) \otimes \pi^*\mathcal{O}_C(2p)$, and we define

$$f\colon X := (x^2 - dy^2 - Gz^2 = 0) \subset \mathbf{P} \longrightarrow C.$$

The zero divisor $Z(G) \subset C$ is reduced. For each $b \in Z(G)$, the fiber X_b has the unique singular point $P_b = [0 : 0 : 1]$. On the chart $z \neq 0$, the equation near P_b is $x^2 - dy^2 = tu$, where t is a uniformizer at b and u is a unit. Hence X is regular at P_b, and it is smooth elsewhere. Since every fiber is integral, $\rho(X/C) = 1$, and $-K_X$ is relatively ample. Thus f is a Mori fiber space, but it is not smooth.

Let $b_a \in C$ be the closed point defined by $U^p - aV^p = 0$ and $P_{b_a} = [0 : 0 : 1] \in X_{b_a}$. After adjoining $\alpha := a^{1/p}$ and $\delta := \sqrt{d}$, work on the charts $V \neq 0$ and $z \neq 0$ and set

$$w := U/V, \qquad s := w - \alpha, \qquad r := x - \delta y, \qquad h := w^p - w.$$

Since h is a unit at $w = \alpha$, putting $q := (x + \delta y)/h$ gives the following completed local equation at the point above P_{b_a}:

$$rq = s^p,$$

which is a rational double point of type A_{p-1}. Hence X is geometrically normal but not geometrically regular.

Set $\xi := \mathcal{O}_{\mathbf{P}}(1)|_X$, let F be the class of a fiber of f over some k-closed point of C, and put $D := \{z = 0\}$. Then $D \sim \xi$ is an integral bisection, and adjunction gives $K_X = -\xi - 2F$. For $\Delta := \left(1 - \frac{2}{p}\right)D$ we have

$$-(K_X + \Delta) = \frac{2}{p}(\xi + pF),$$

which is nef because $\mathcal{E} \otimes \mathcal{O}_C(p)$ is globally generated. Moreover, (X, Δ) is geometrically klt: over $\overline{k}$, the divisor D splits into two disjoint smooth sections avoiding the singular locus of $X_{\overline{k}}$. In particular, all the hypotheses of Proposition 3.1 except smoothness hold, whereas its conclusion fails.

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